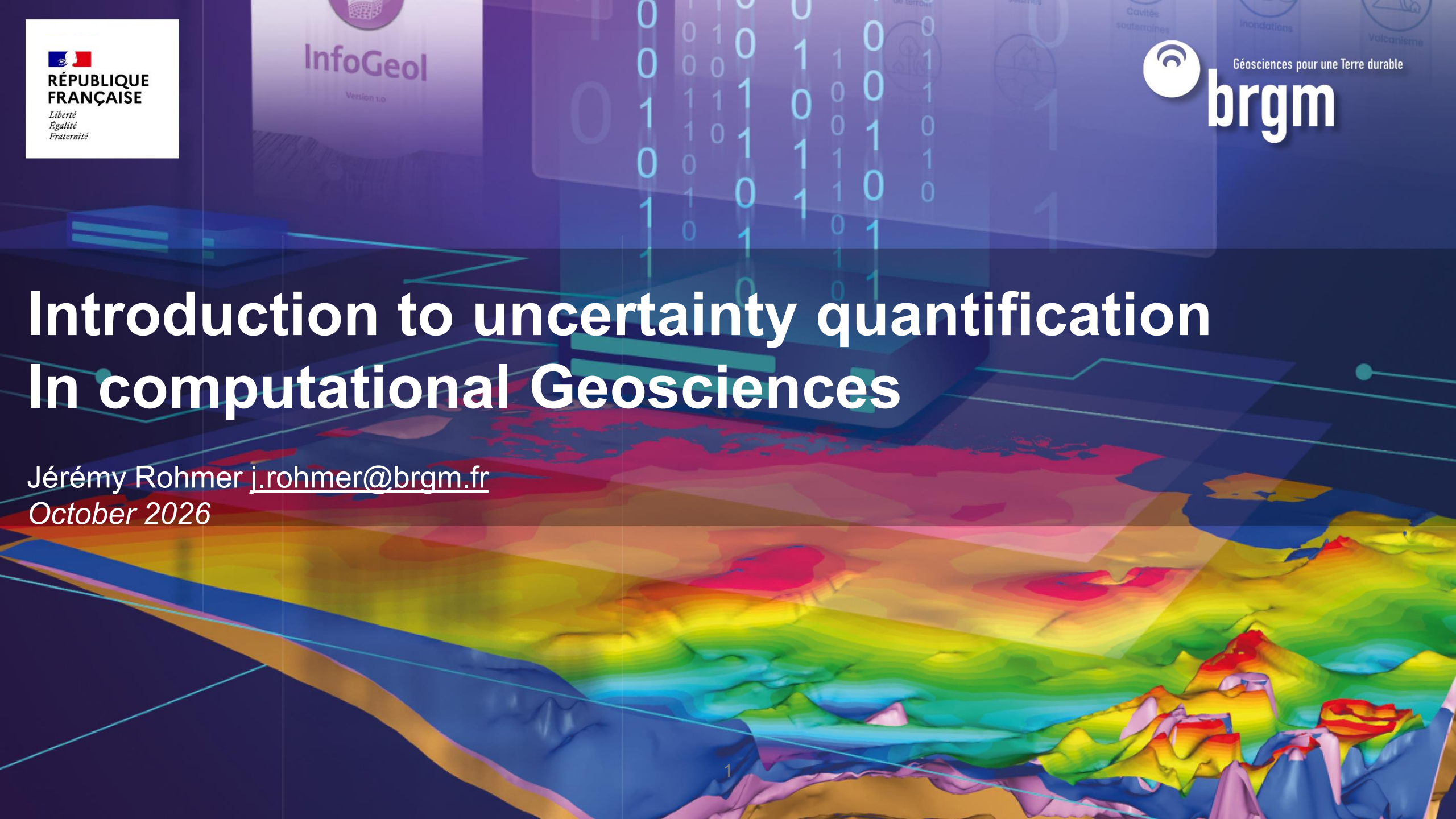


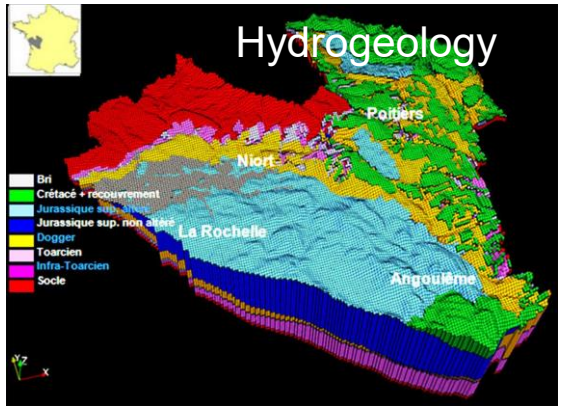
Introduction to uncertainty quantification In computational Geosciences

Jérémy Rohmer j.rohmer@brgm.fr

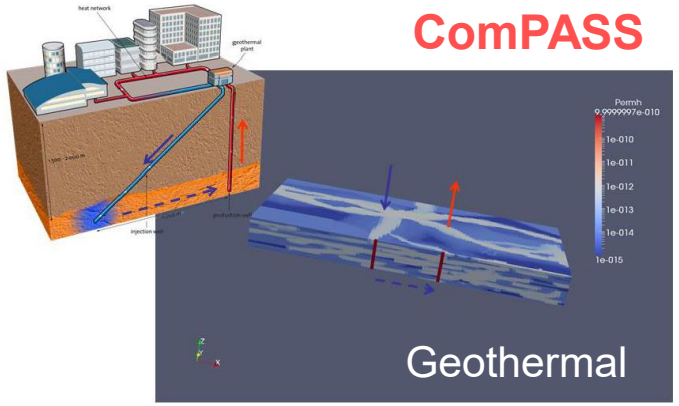
October 2026

Key role of numerical models

Softwares developed w.r.t. BRGM activities



MARTHE

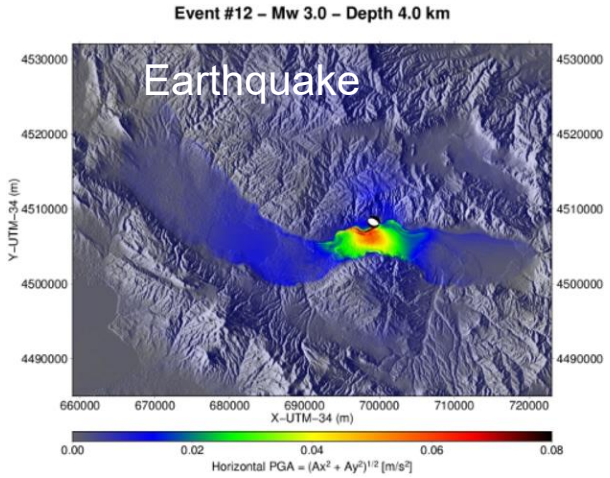


ComPASS

Computational Model

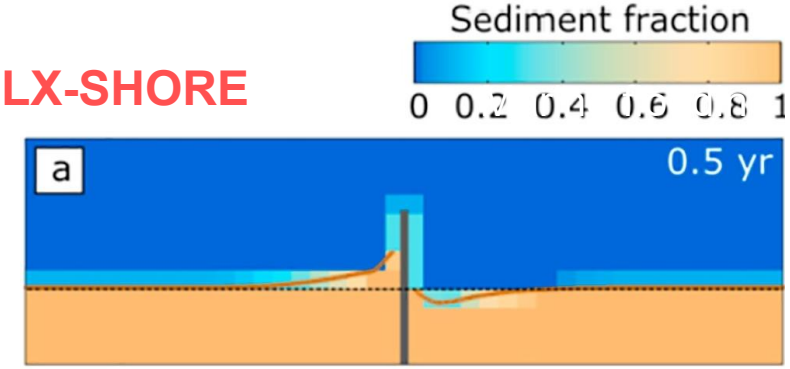
e.g. urban planning, crisis management, natural resource exploitation, etc.

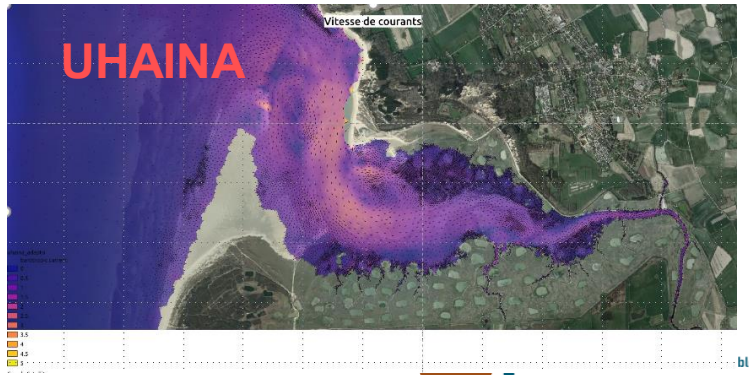
Support for decision making



EFISPEC3D

LX-SHORE





Non-exhaustive list

Key role of numerical models

Computational Model

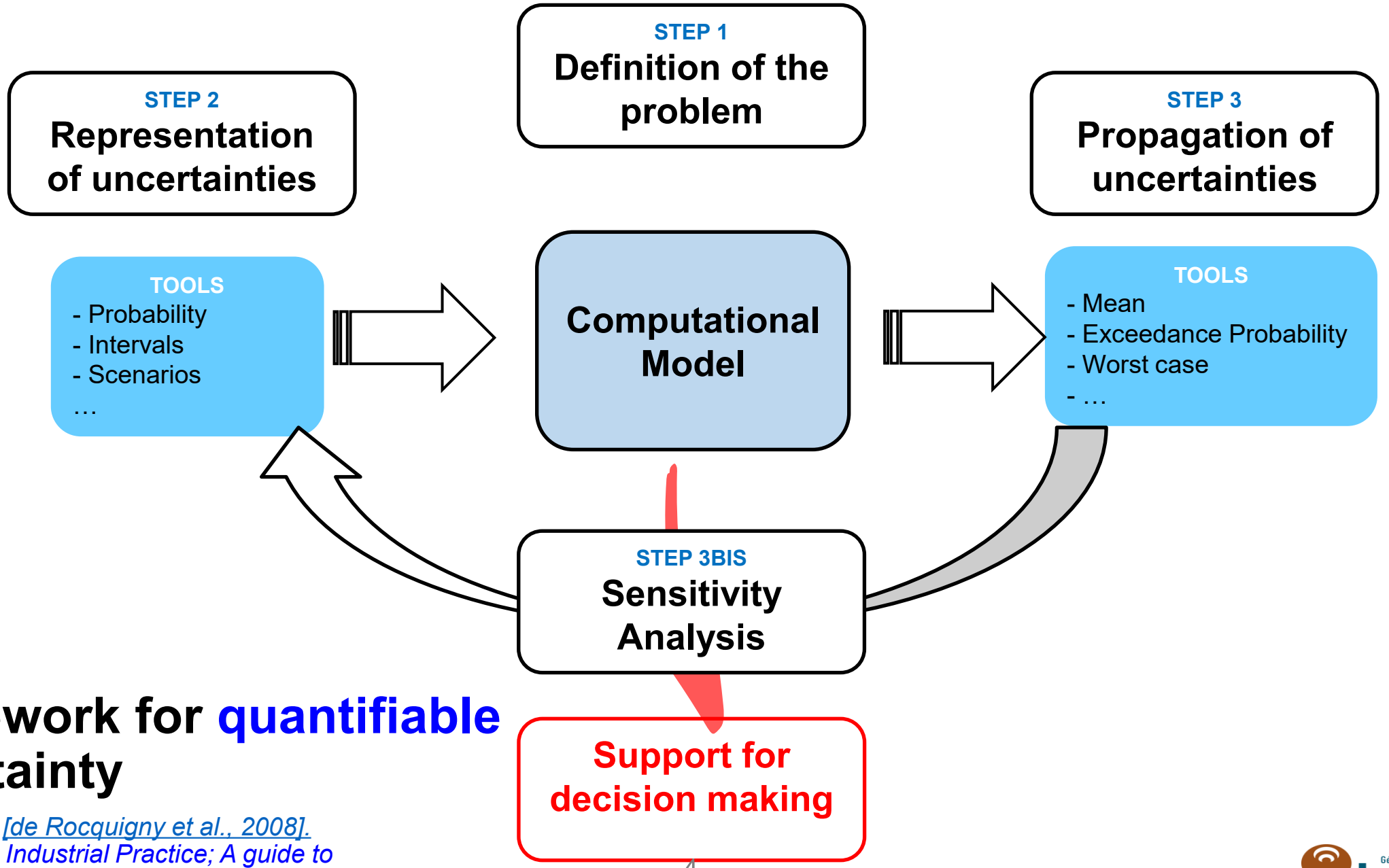
Key questions:

- **Confidence** in the numerical results?
- How to **reduce uncertainty**?



Support for decision making

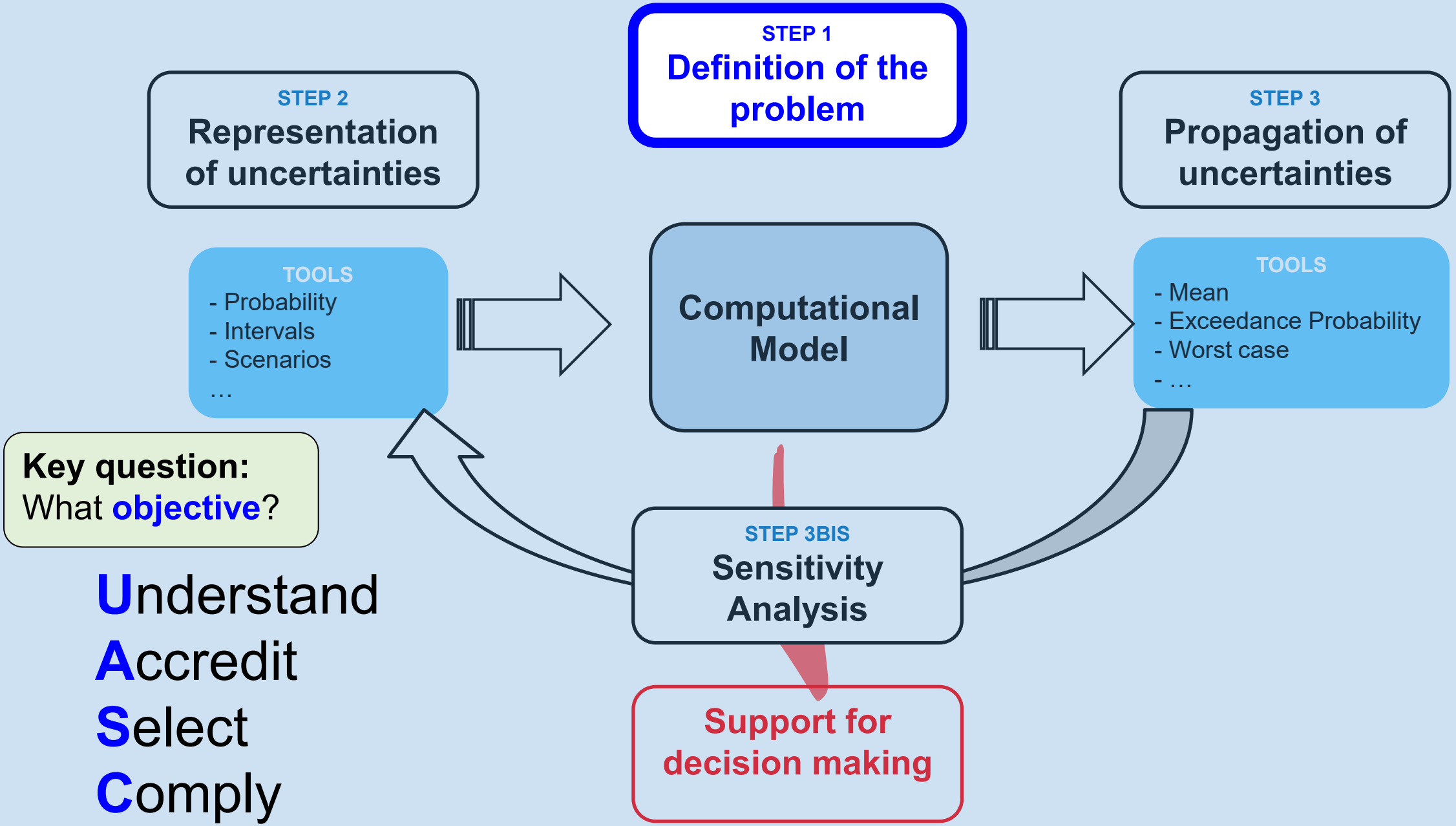
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Framework for quantifiable uncertainty

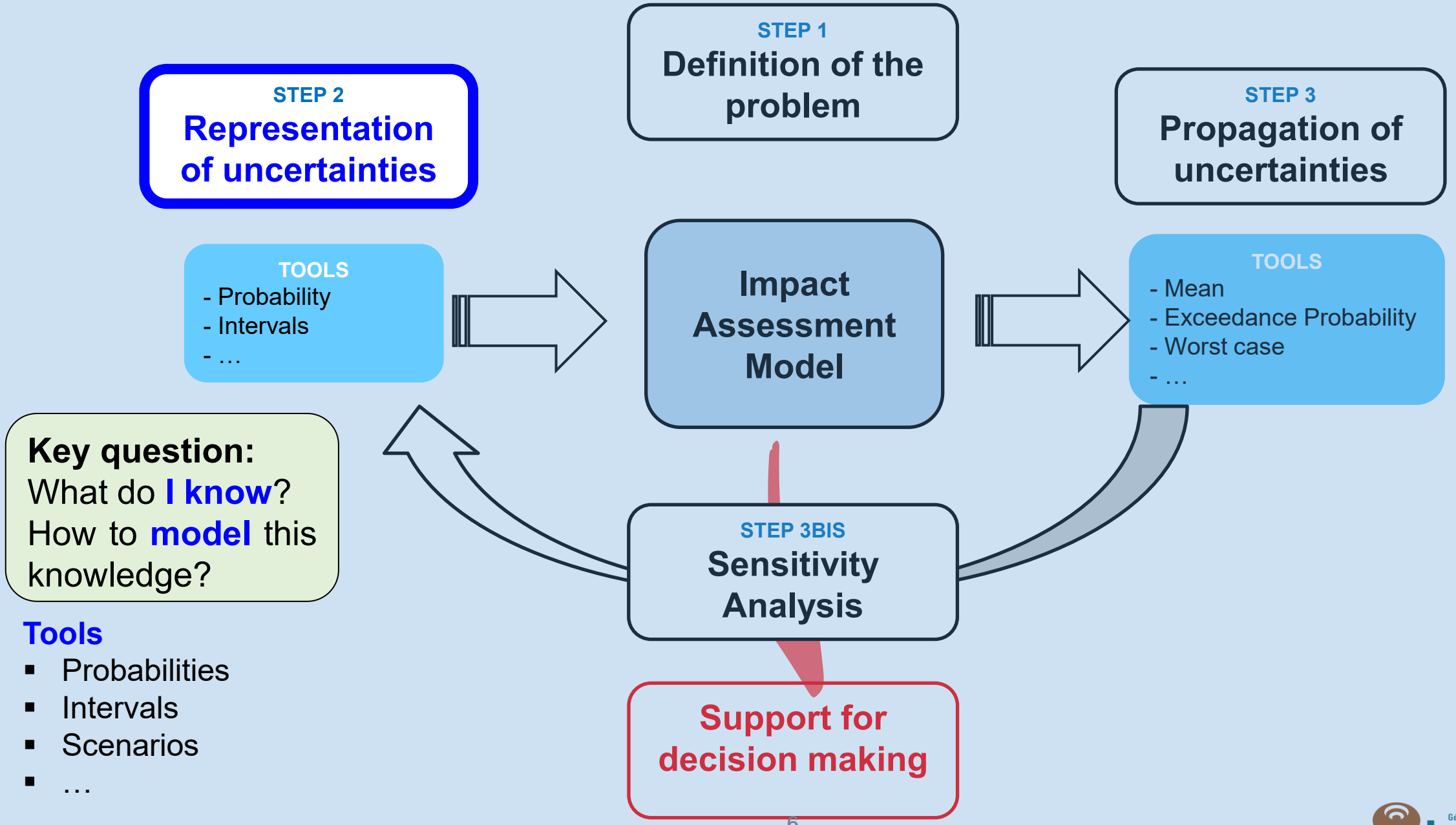
Adapted from [de Rocquigny et al., 2008]. Uncertainty in Industrial Practice; A guide to Quantitative Uncertainty Management, Wiley

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Understand
Accredit
Select
Comply

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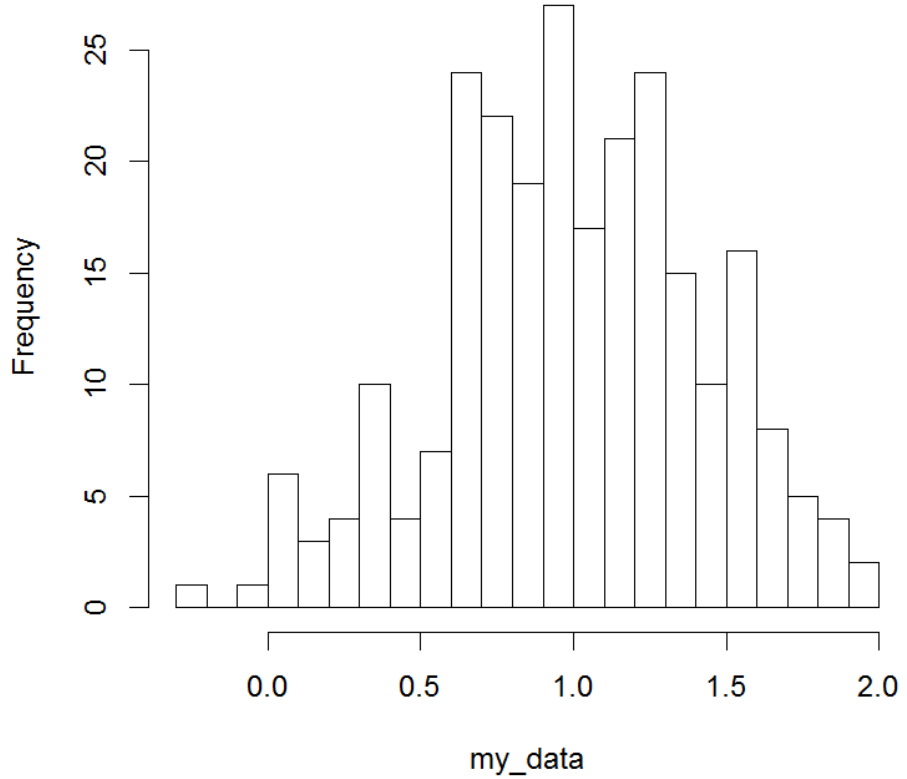
Key question:
What do **I know**?
How to **model** this knowledge?

Tools

- Probabilities
- Intervals
- Scenarios
- …

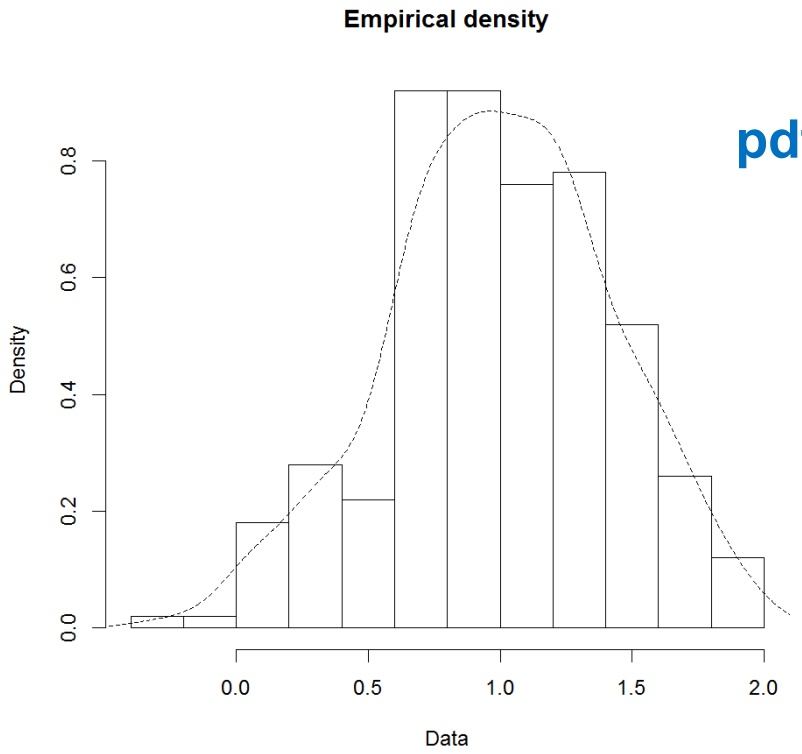
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The tools of probability theory



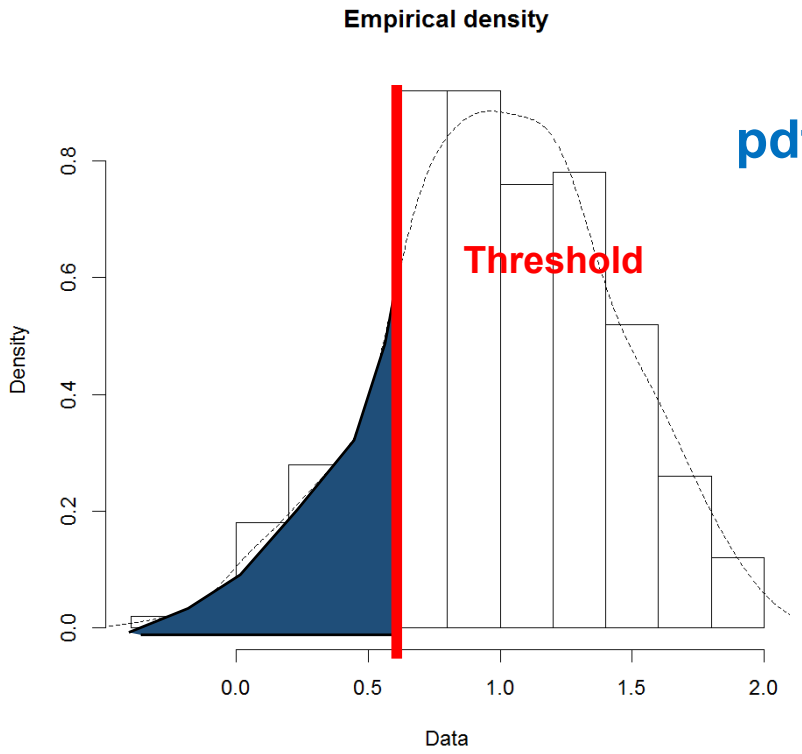
250 observations

The tools of probability theory



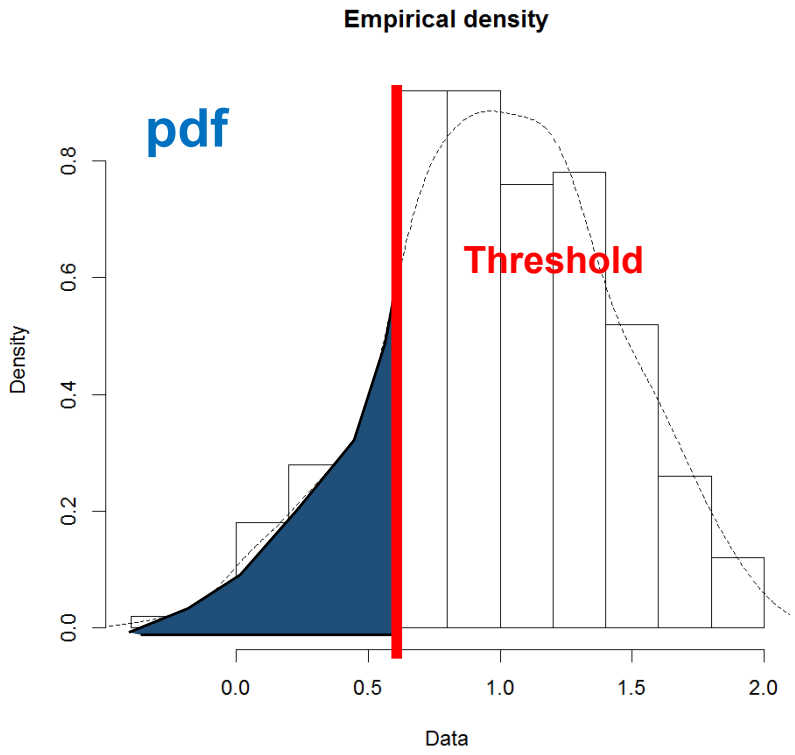
pdf: probability density function

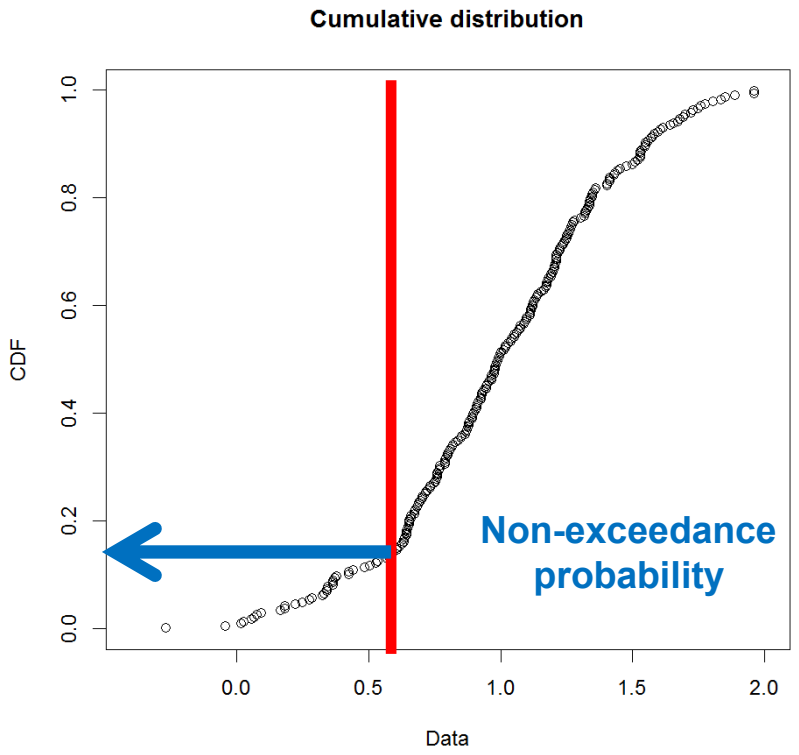
The tools of probability theory



pdf: probability density function

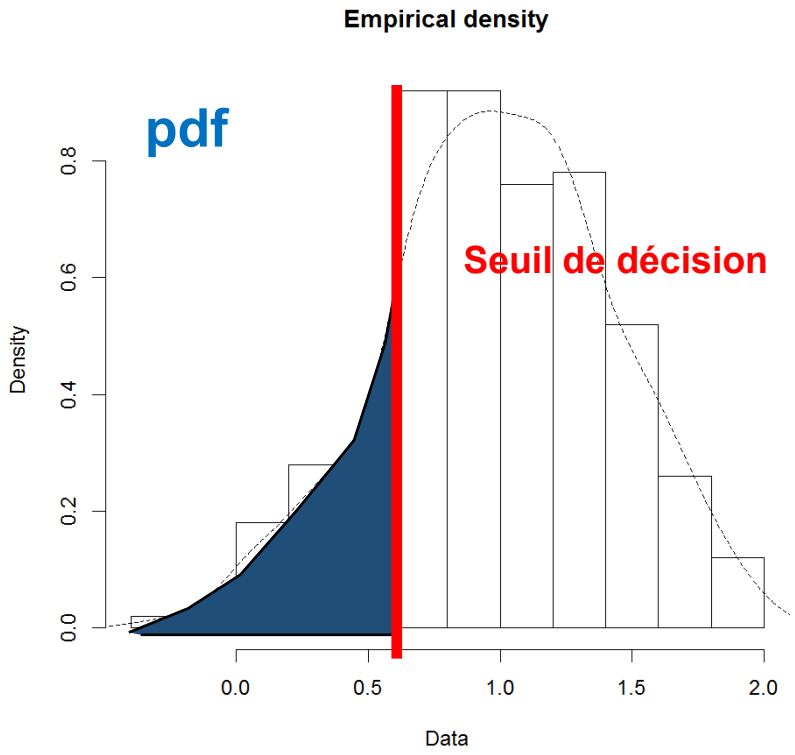
The tools of probability theory

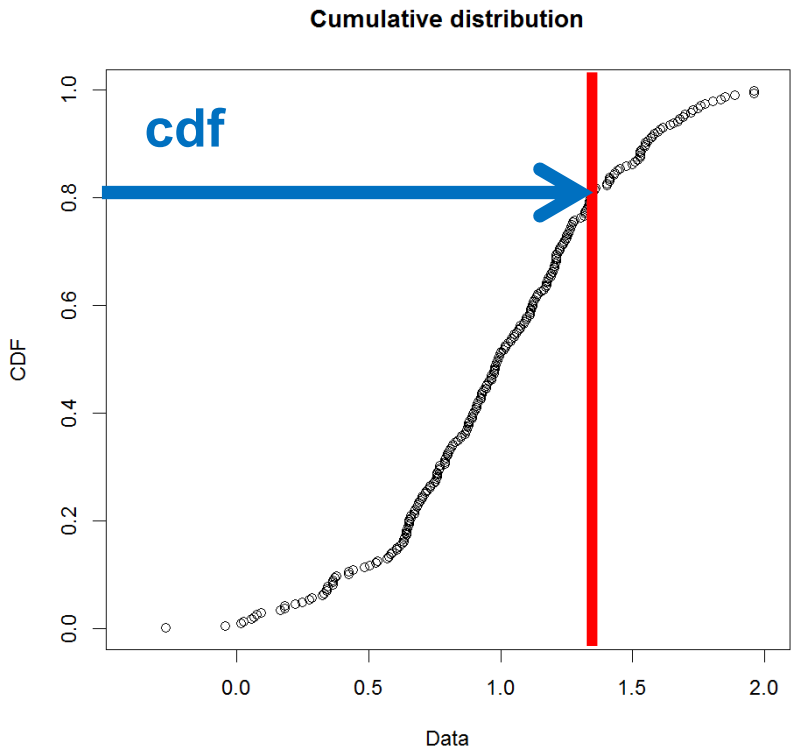




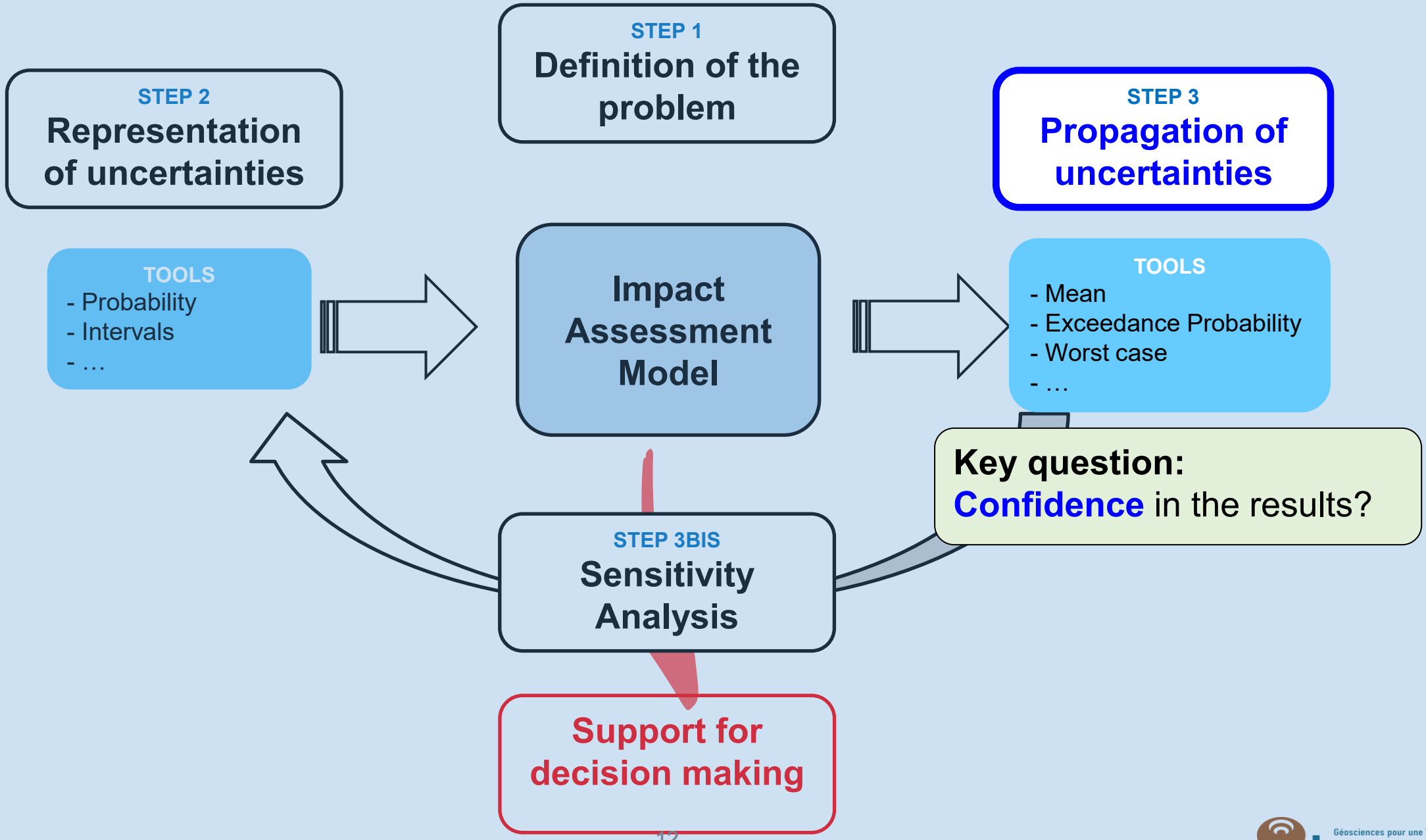
cdf: cumulative distributuon function

The tools of probability theory



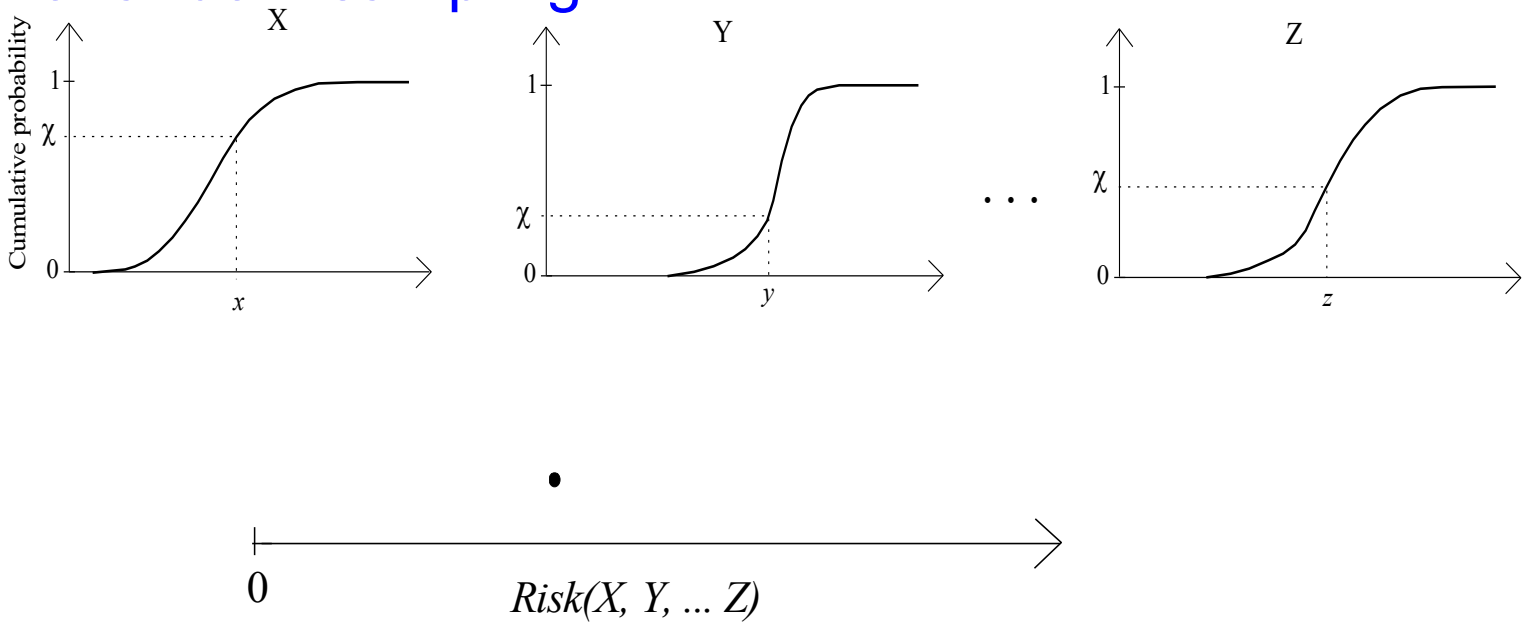


Percentile at 80%


STEP 1
Definition of the problem
STEP 2
Representation of uncertainties
STEP 3
Propagation of uncertainties
TOOLS
- Probability
- Intervals
- …
Impact Assessment Model
TOOLS
- Mean
- Exceedance Probability
- Worst case
- …
Key question:
Confidence in the results?
STEP 3BIS
Sensitivity Analysis
Support for decision making

Depending on the objective, different approaches…

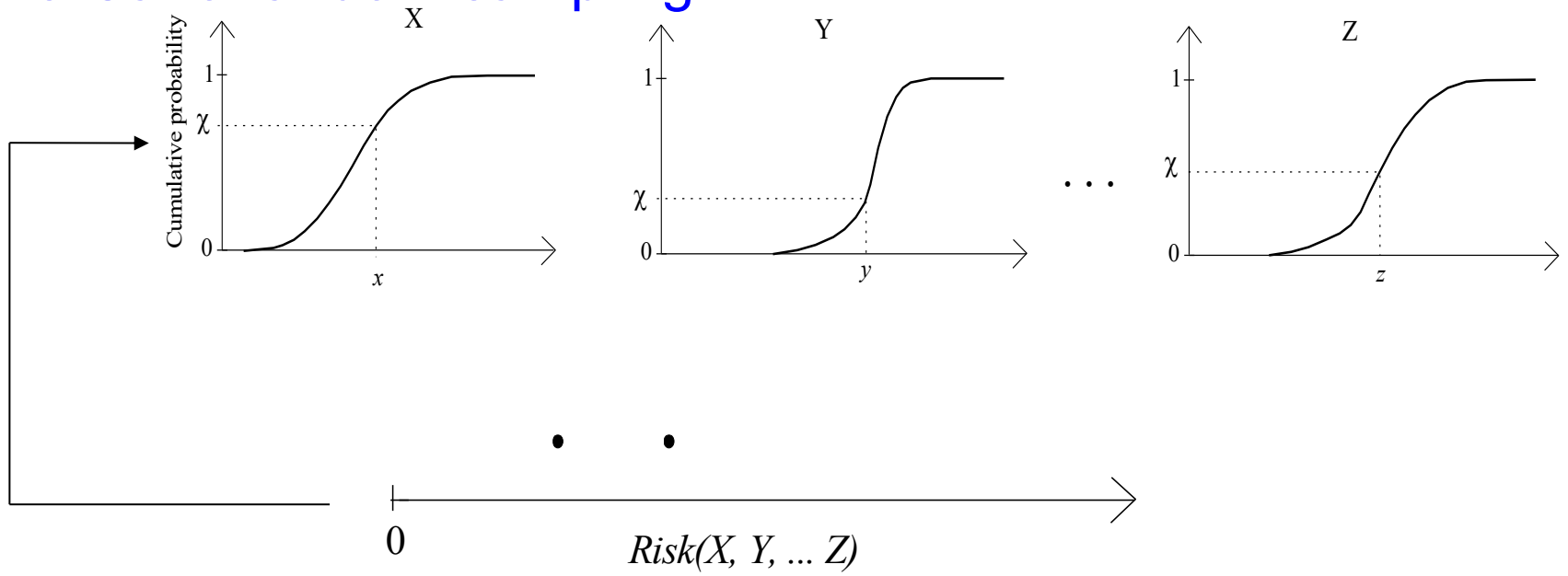
- Monte-Carlo random sampling



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Depending on the objective, different approaches…

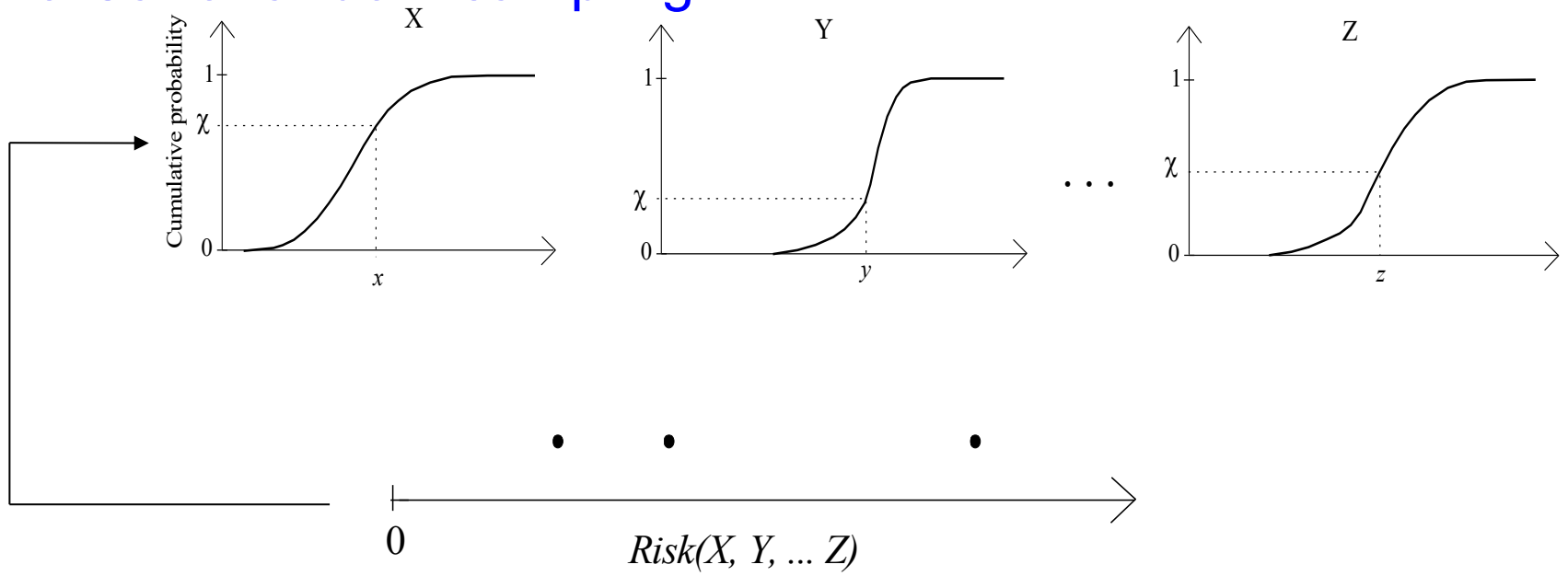
- Monte-Carlo random sampling



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Depending on the objective, different approaches…

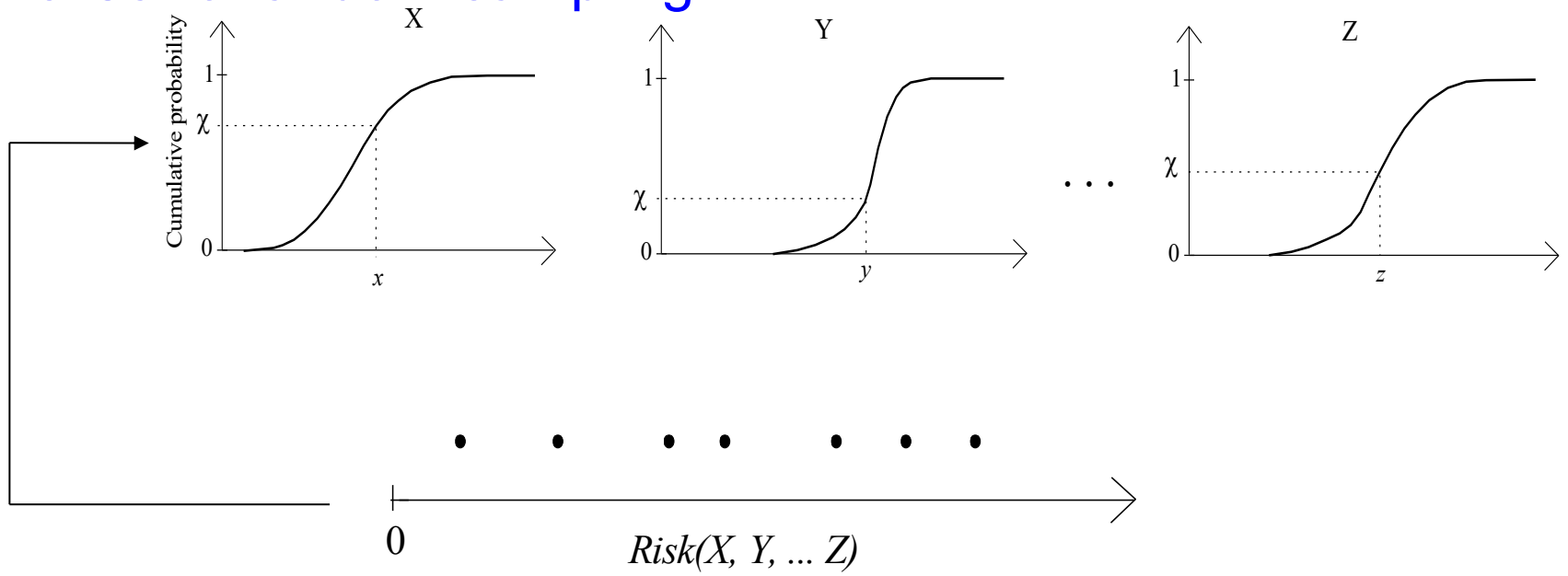
- Monte-Carlo random sampling



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Depending on the objective, different approaches…

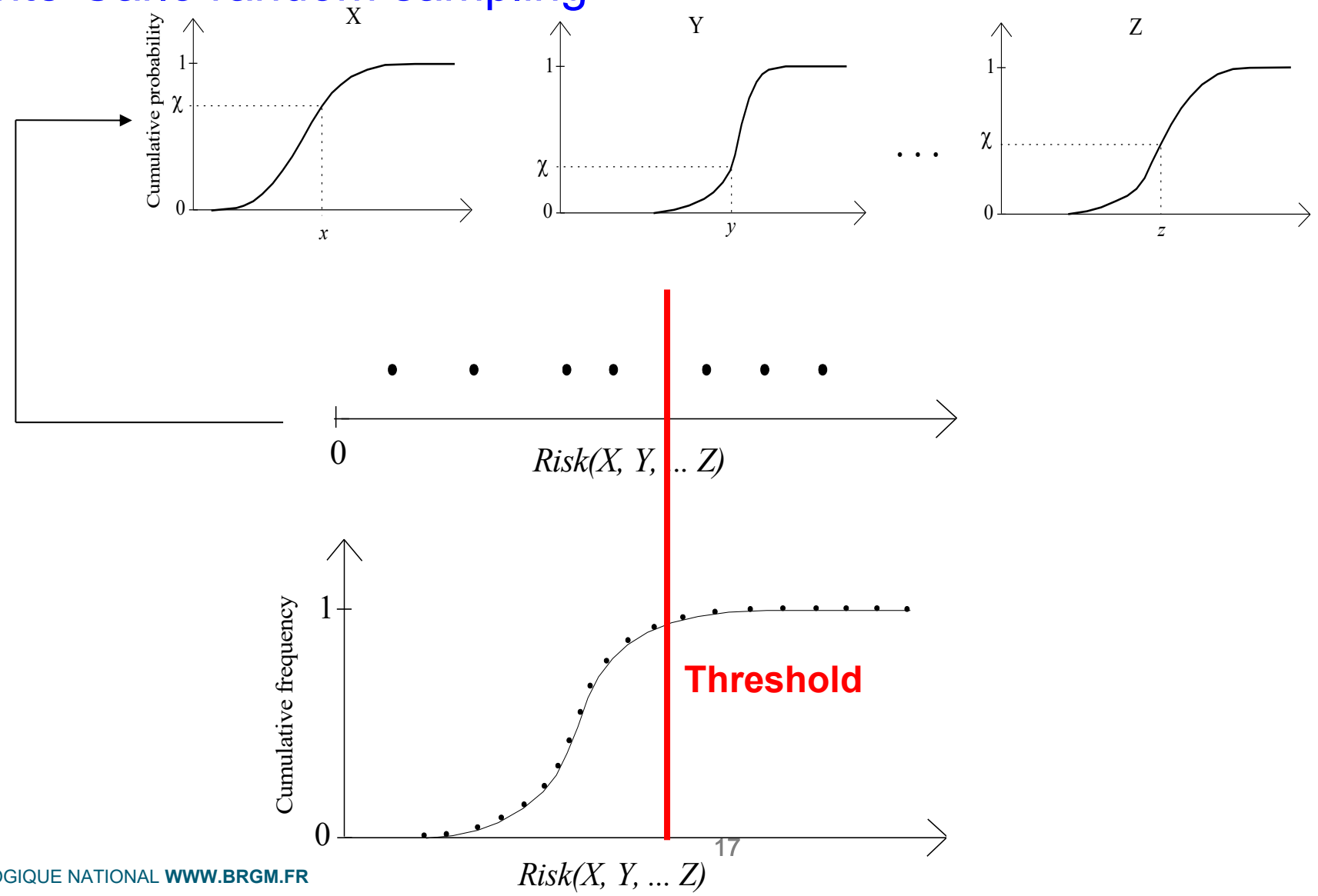
- Monte-Carlo random sampling



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Depending on the objective, different approaches…

- Monte-Carlo random sampling



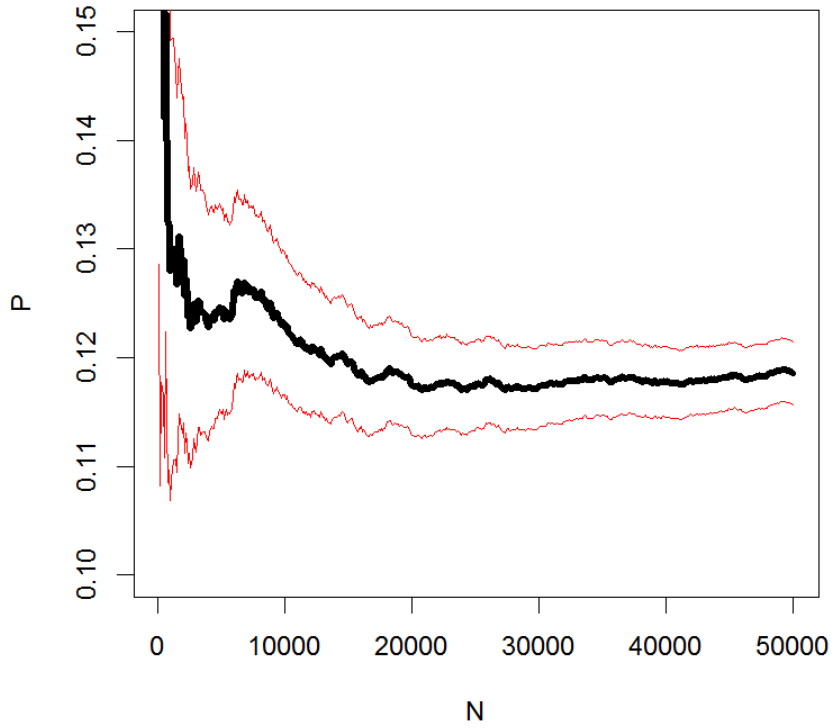
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Keystone: the convergence

$$P = \frac{1}{N}\left(\sum_{i=1}^{N} I_{\{FS<1\}}\right)$$

$I_{\{A\}} = 1$ If A is true

N = number of samples



Rule of thumb if P is low

$$N \sim \frac{1}{P.V}$$

Example P=10^{-4} with error V = 10%,

N=1/(10^{-4}x0.1^2)=10^6

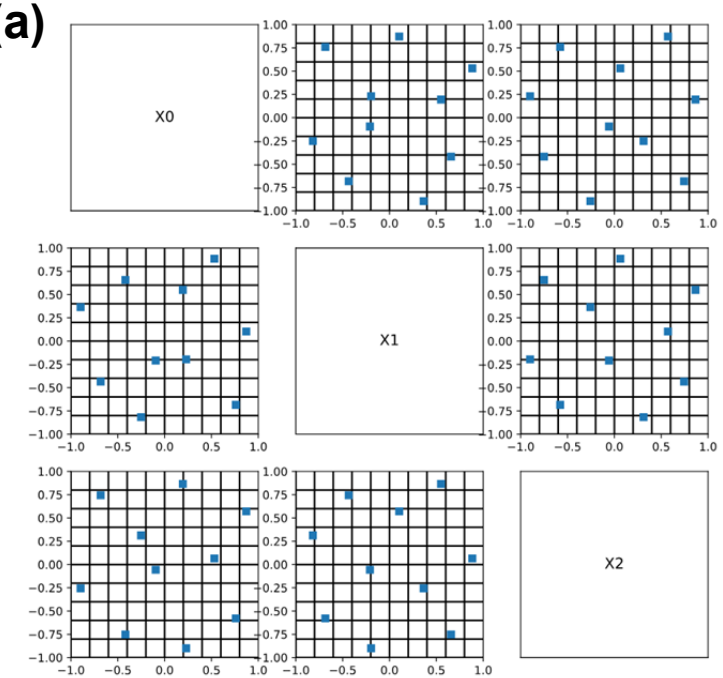
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Depending on the objective, different approaches…

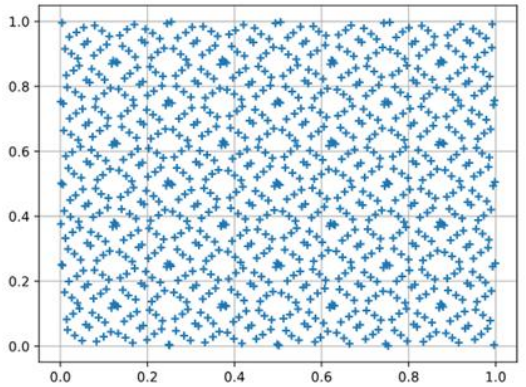
- Monte-Carlo random sampling

- ‘Accelerated’ Monte Carlo sampling
  - Latin Hypercube Sampling (&variants) **(a)**
  - Low discrepancy random sequences **(b)**
  - Importance sampling
  - Multi-level sampling **(c)**
  - …

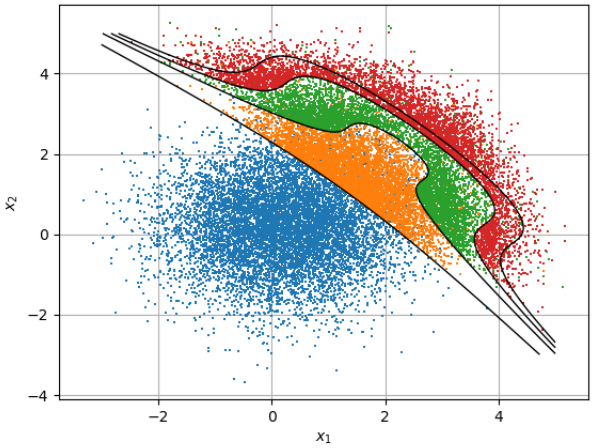
(a)



(b)

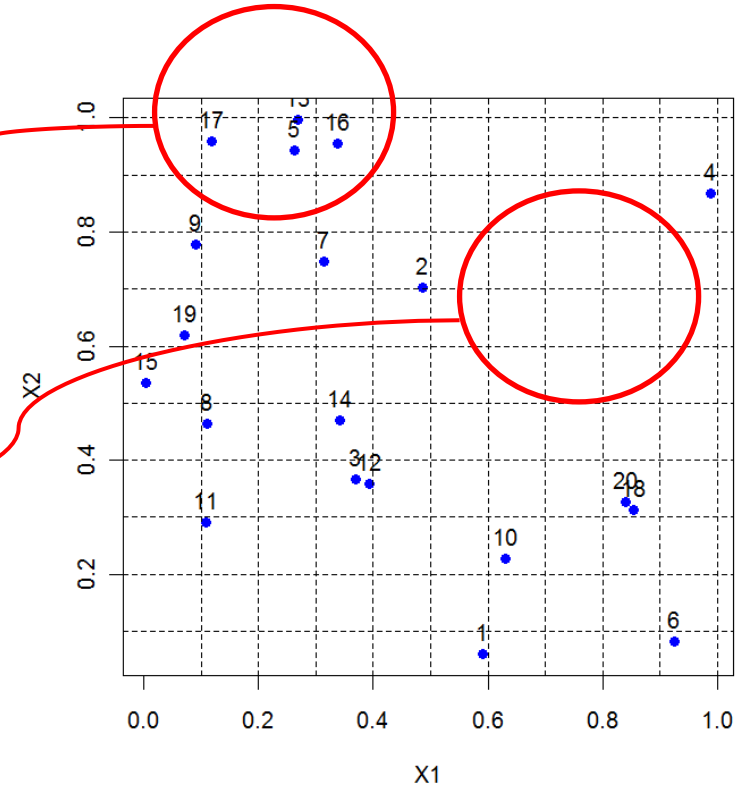


(c)

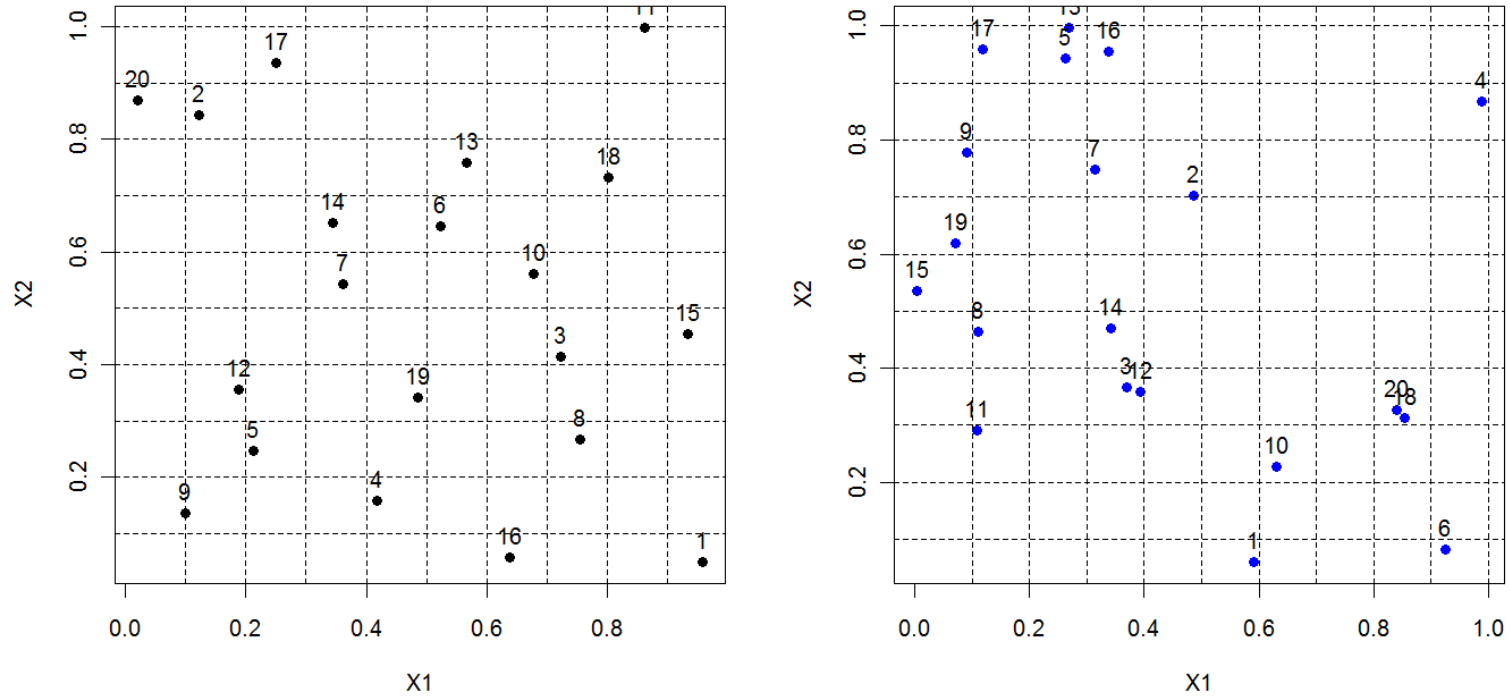


Clustering effect

« Hole »



Uniform sampling



Latin Hypercube Sampling

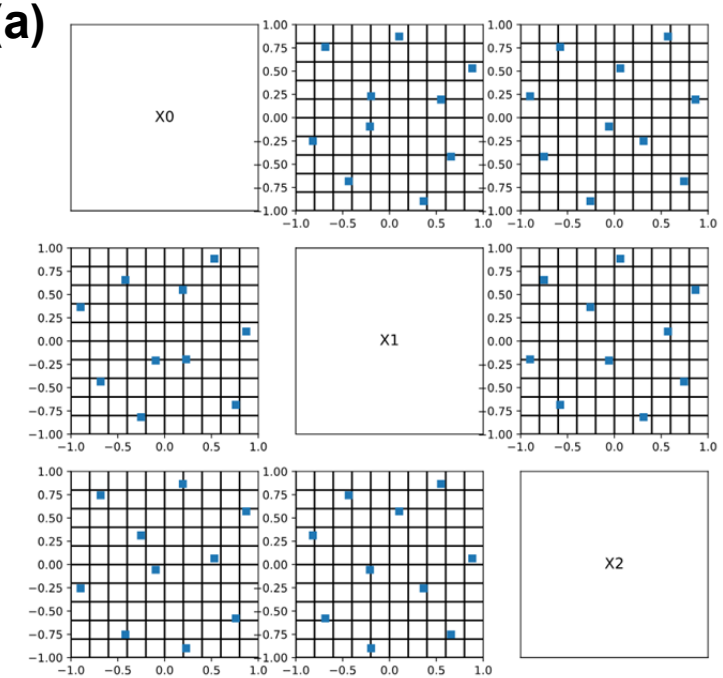
[McKay et al. 1979]

https://bertcarnell.github.io/lhs/articles/lhs_basics.html

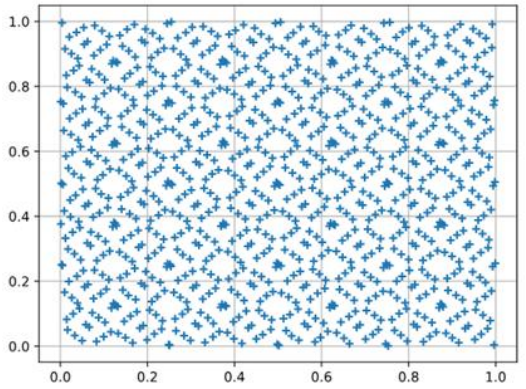
Depending on the objective, different approaches…

- Monte-Carlo random sampling
- ‘Accelerated’ Monte Carlo sampling
  - Latin Hypercube Sampling (&variants) **(a)**
  - Low discrepancy random sequences **(b)**
  - Importance sampling
  - Multi-level sampling **(c)**
  - …
- Reliability methods
  - FORM/SORM
  - Monotonic Reliability Method
- Interval computation and optimisation

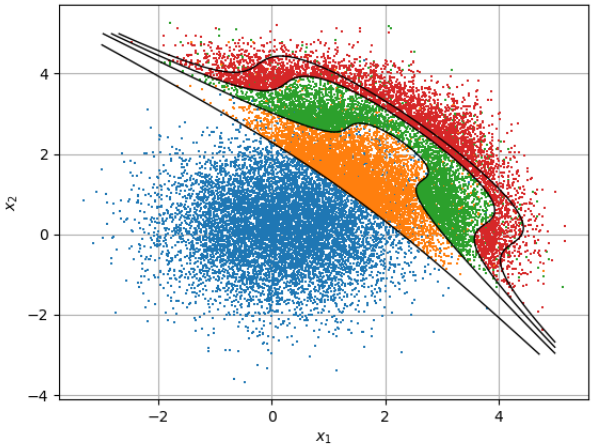
(a)

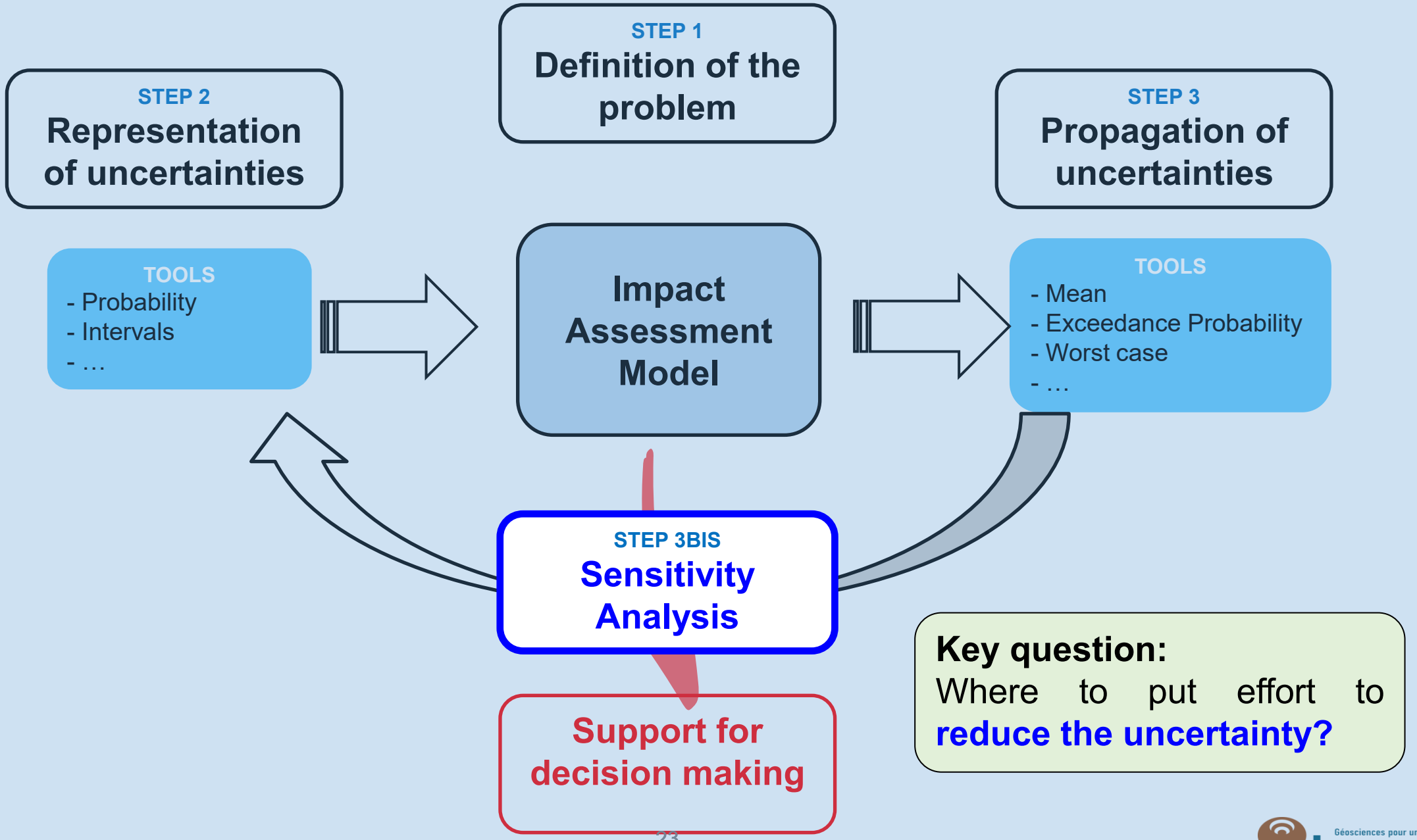


(b)



(c)




STEP 1
Definition of the problem
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- …
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- Worst case
- …
STEP 3BIS
Sensitivity Analysis
Support for decision making
Key question:
Where to put effort to reduce the uncertainty?

"For every dollar that is spent trying to quantify uncertainty, we should spend 10 dollars collecting and analyzing data that would *reduce uncertainty*".

[Gail Atkinson] (2004 World Conference on Earthquake Engineering)

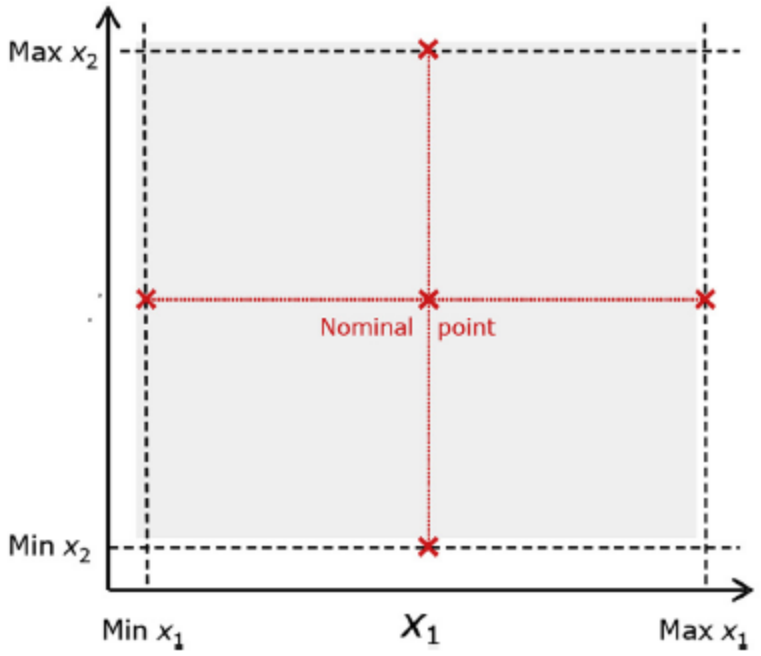
Two possible approaches

- Local
  - One parameter is varied ‘one at a time’ OAT
  - Quantify a deviation from a reference solution

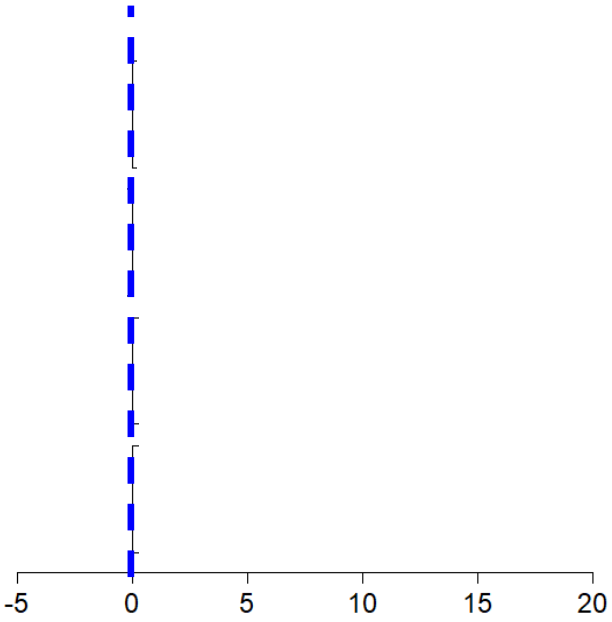
- Global
  - All parameters are varied all to together
  - Account for probabilistic information
  - Account for interactions

Local sensitivity analysis « OAT »

input



output

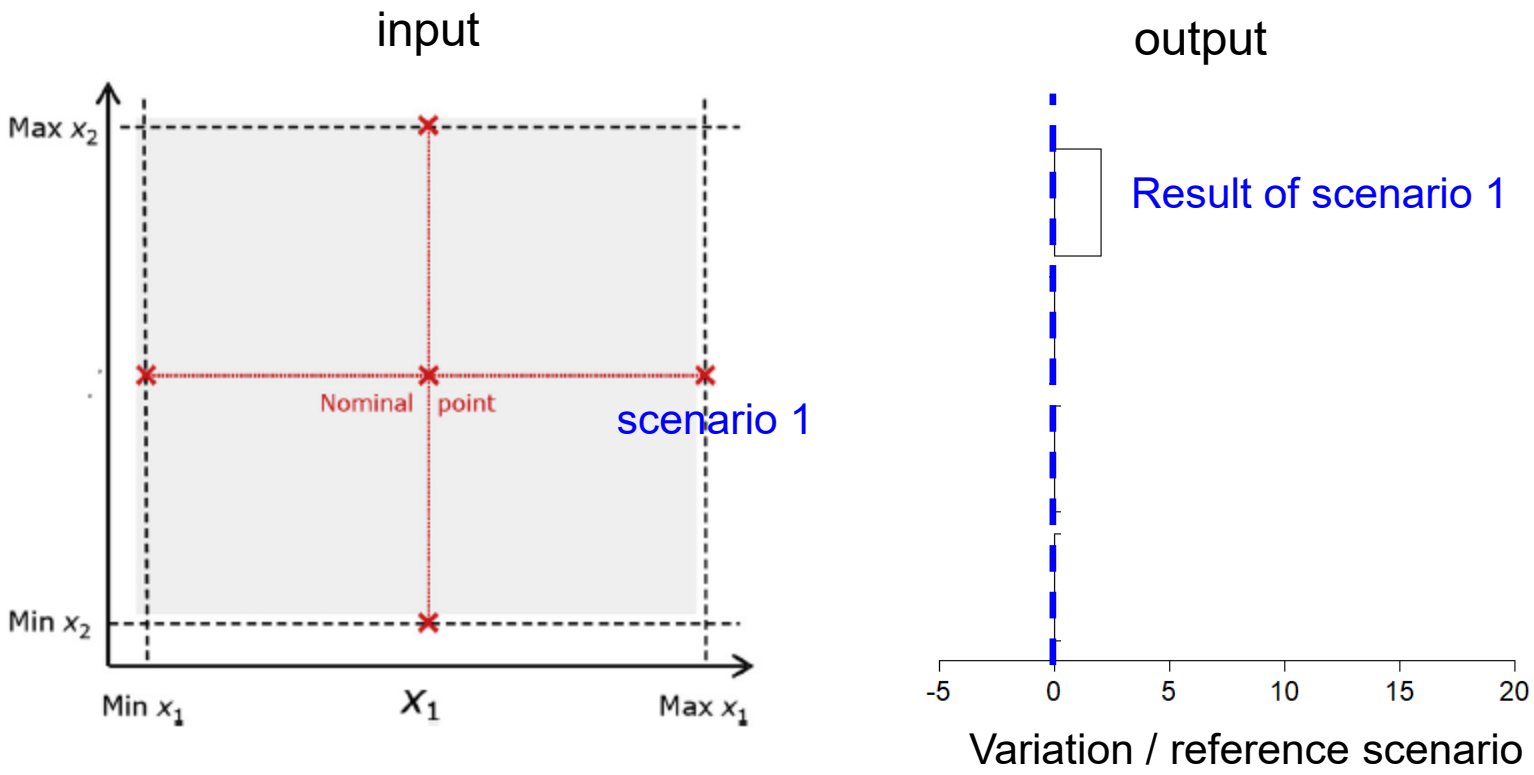


Variation / reference scenario

= nominal point

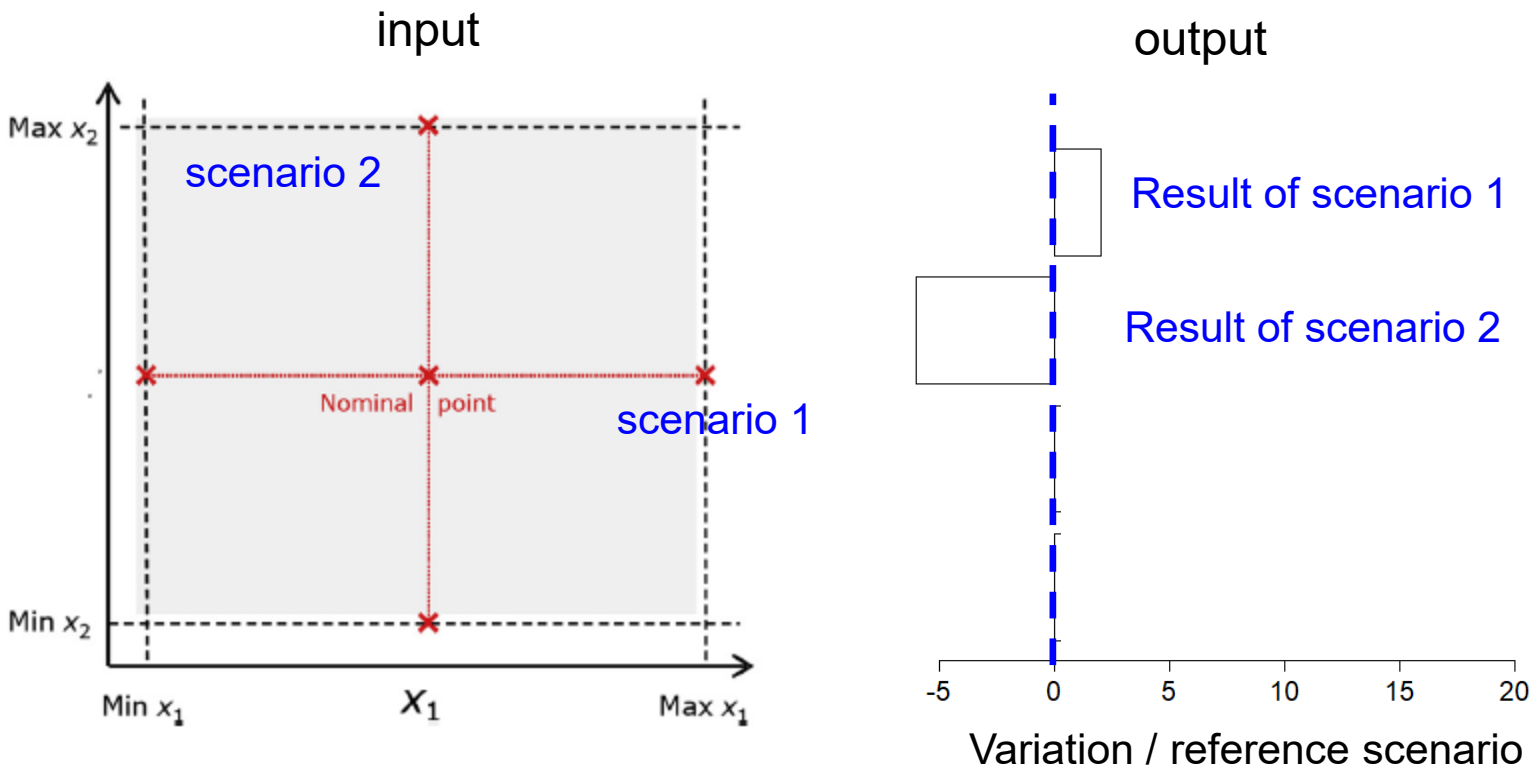
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Local sensitivity analysis « OAT »



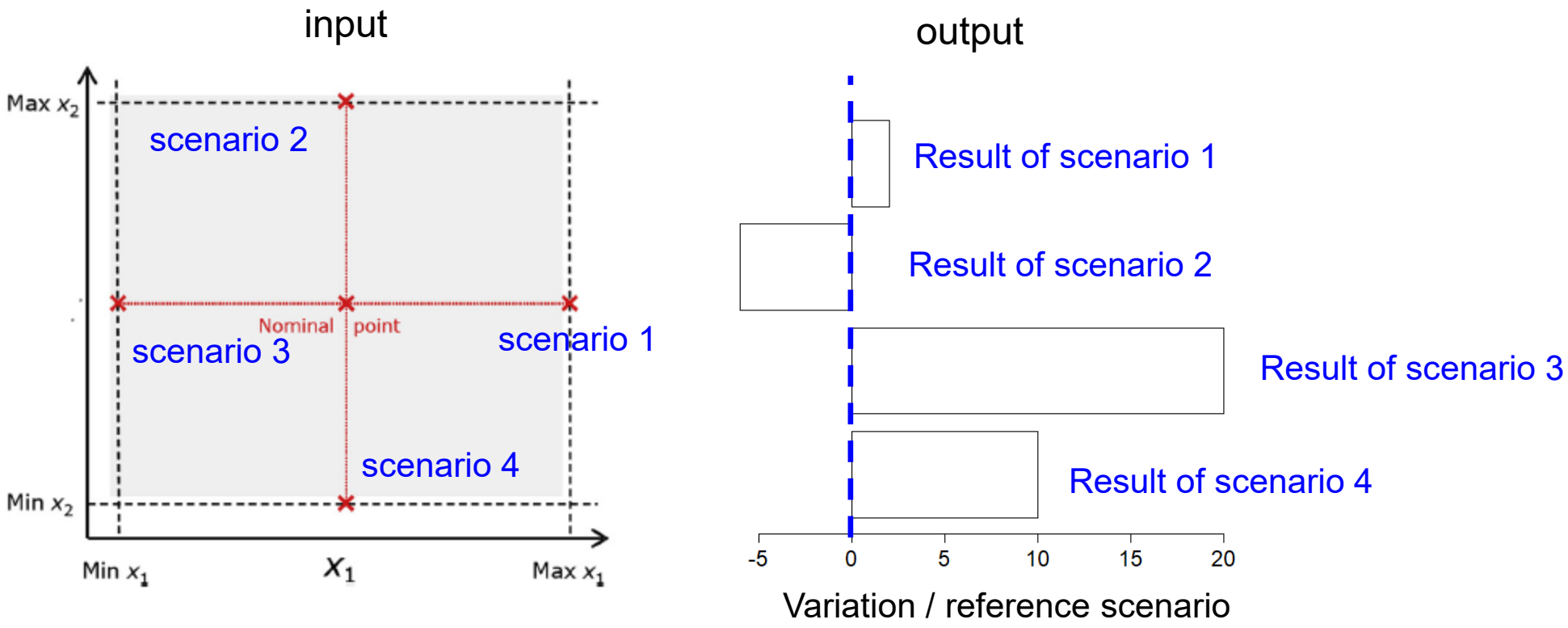
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Local sensitivity analysis « OAT »



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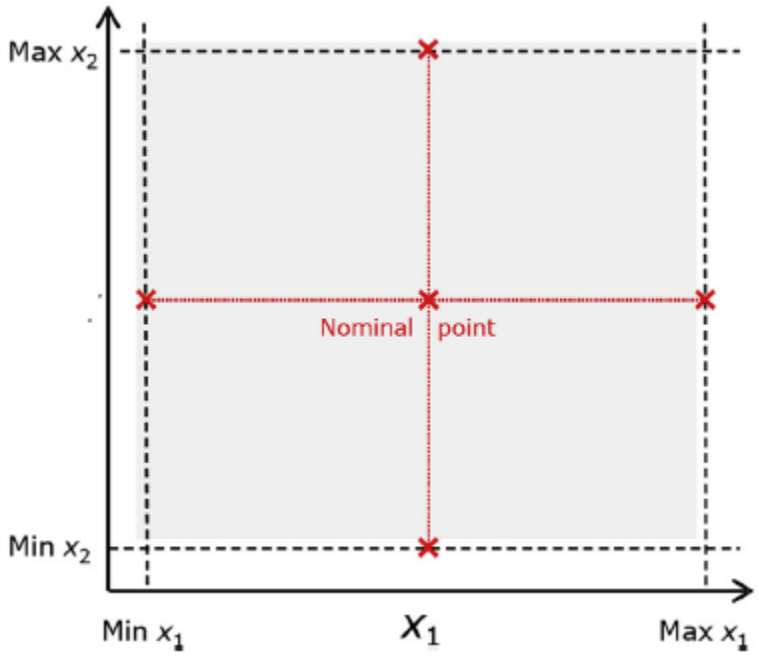
Local sensitivity analysis « OAT »



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Two possible approaches

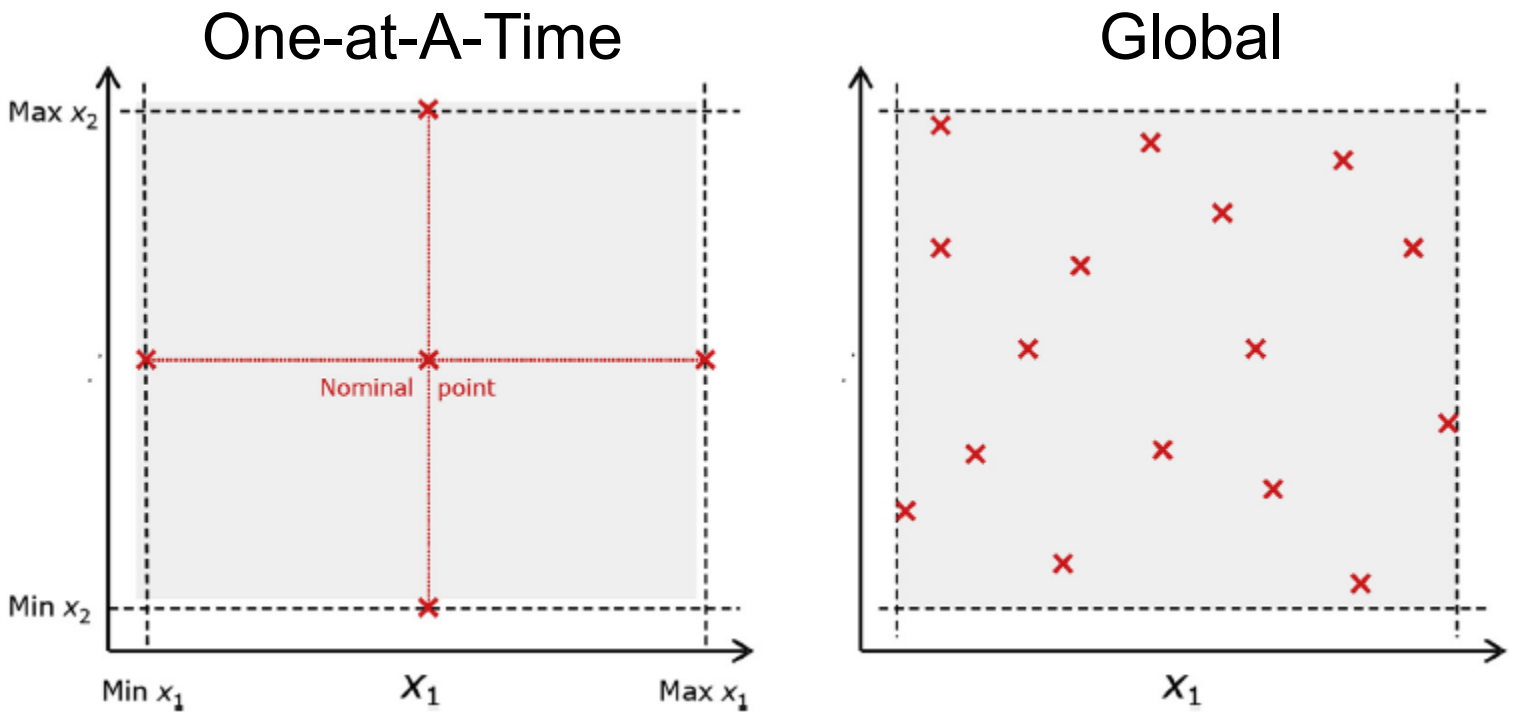
One-at-A-Time





- Dependence on the reference
- No exploration of interactions
- No accounts for probabilistic information

Two possible approaches



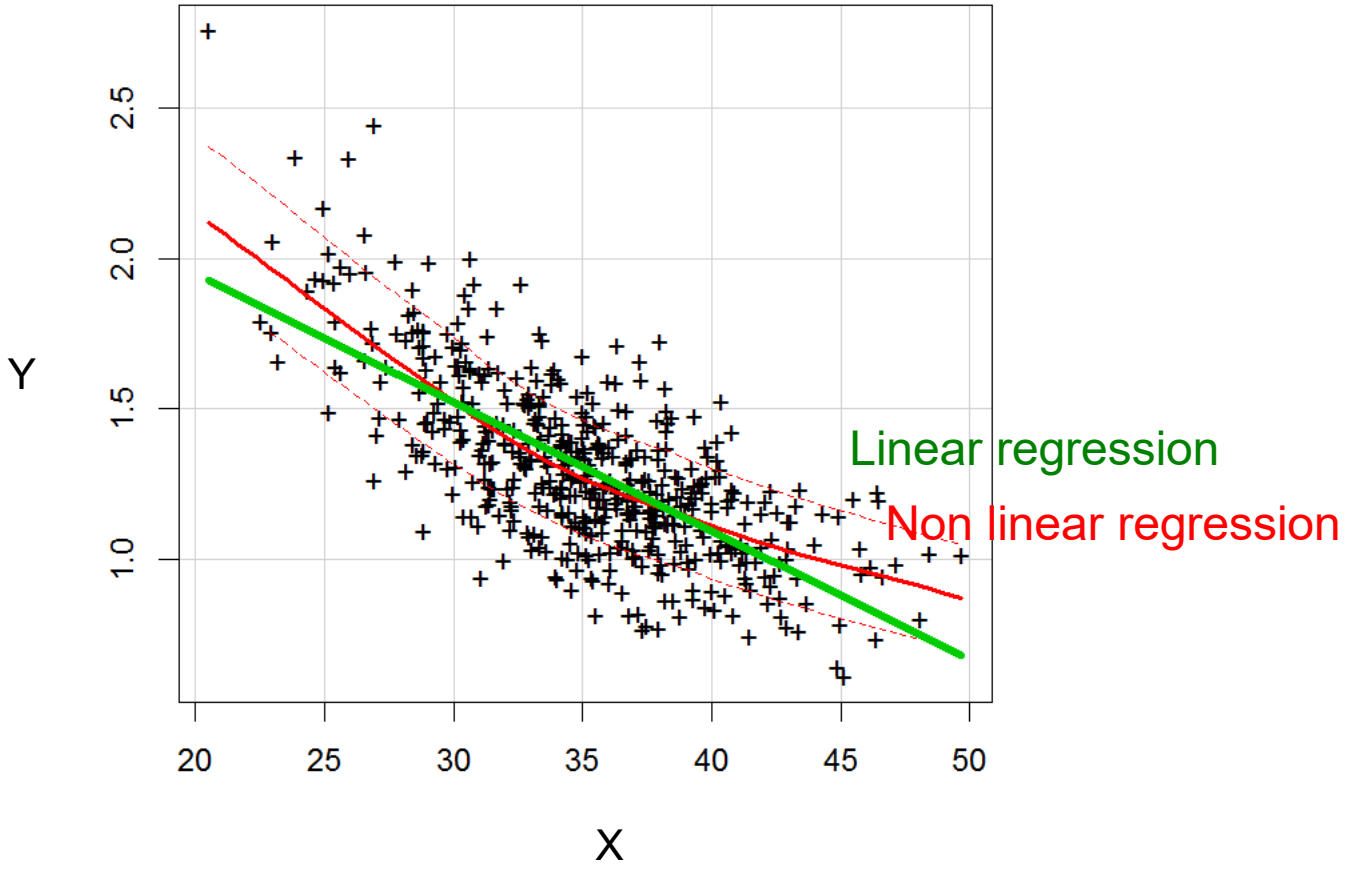


- No reference
- Explore interactions
- Accounts for probabilistic information

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Global approach 1: correlation analysis

500 randomly sampled data



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Difference types of correlation

https://analyse-it.com/learn/pearson-spearman-or-kendall



Global approach 2: decomposition of variance

[Saltelli et al., 2008]

- Objective:
  - Explore the influence of inputs X_i (i=1..d) on the output y;
  - Over the whole range of variation of X_i;
  - With account of interactions
  - Model-free

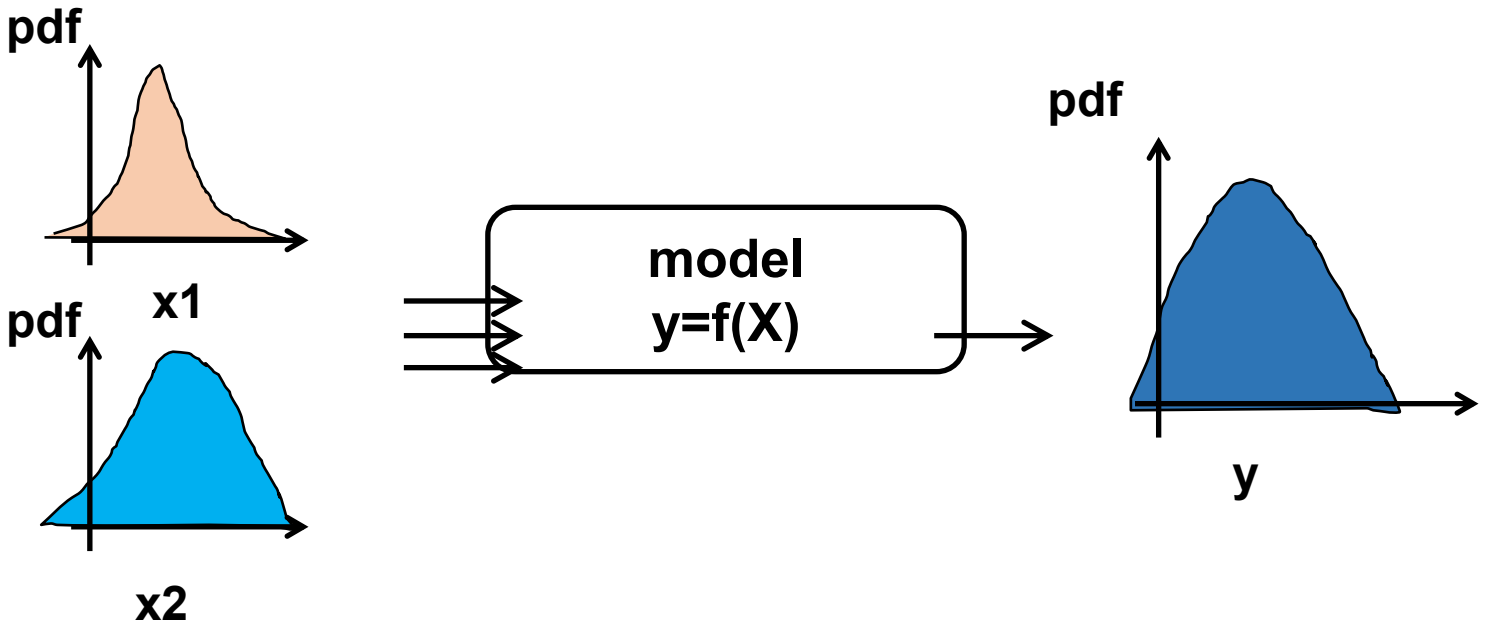
- Principle:
  - Estimate the variance reduction of Y if I knew the true value x^*_i of X_i
  - The input leading to the largest reduction has the largest influence

- Assumptions
  - Variance(Y) ~ measure of uncertainty
  - X_i are independent

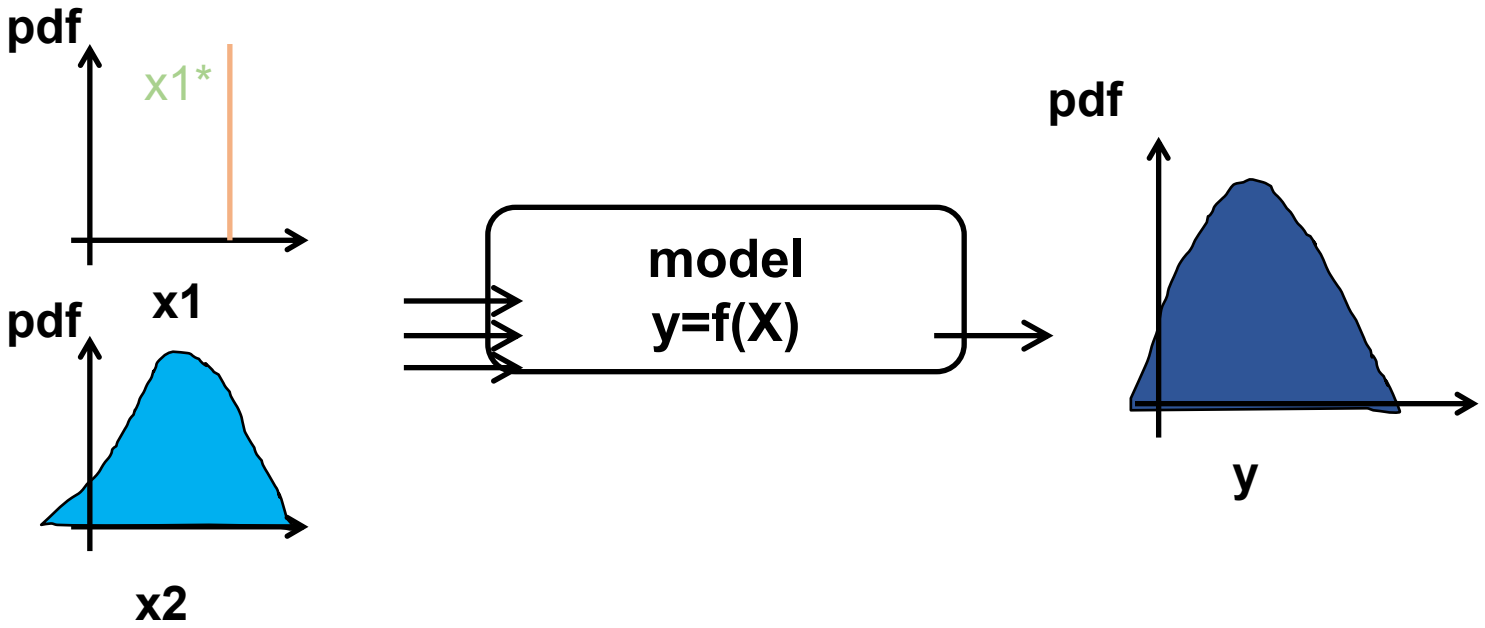
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Inputs

Output

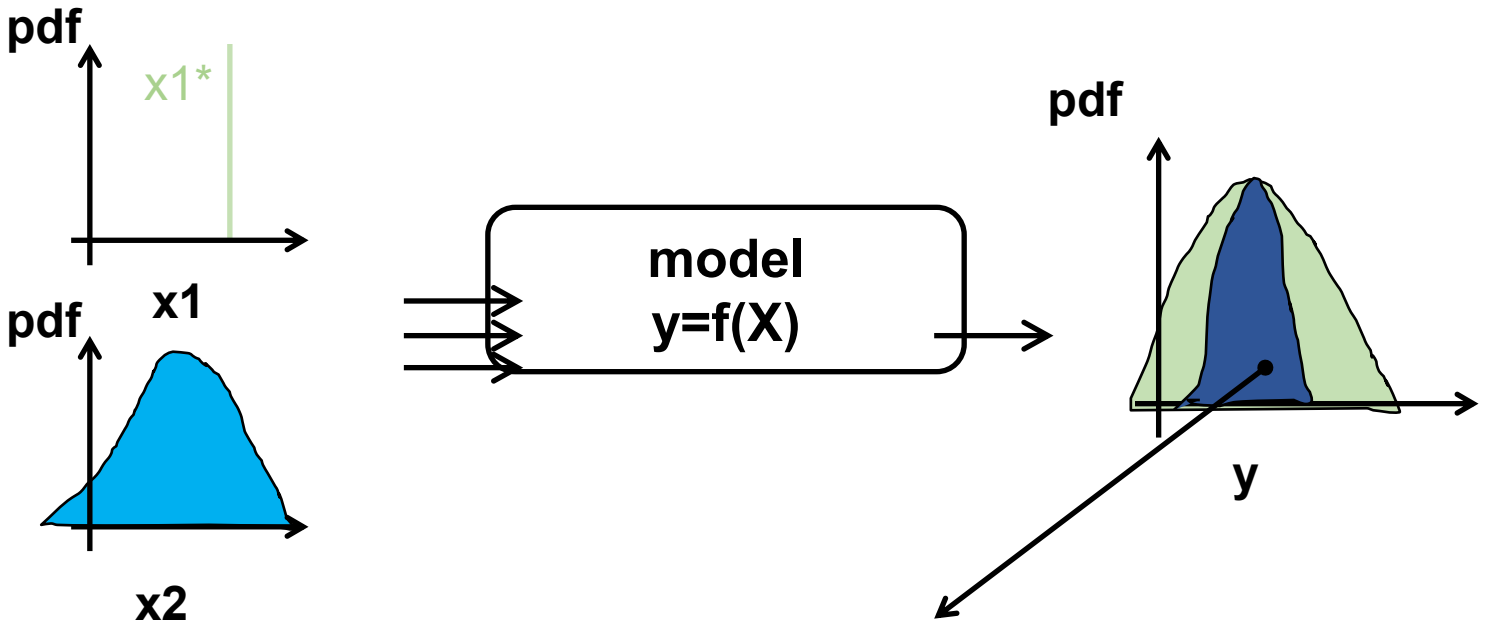


If the true value x1* was known the variance V(Y) would be reduced by V(1)



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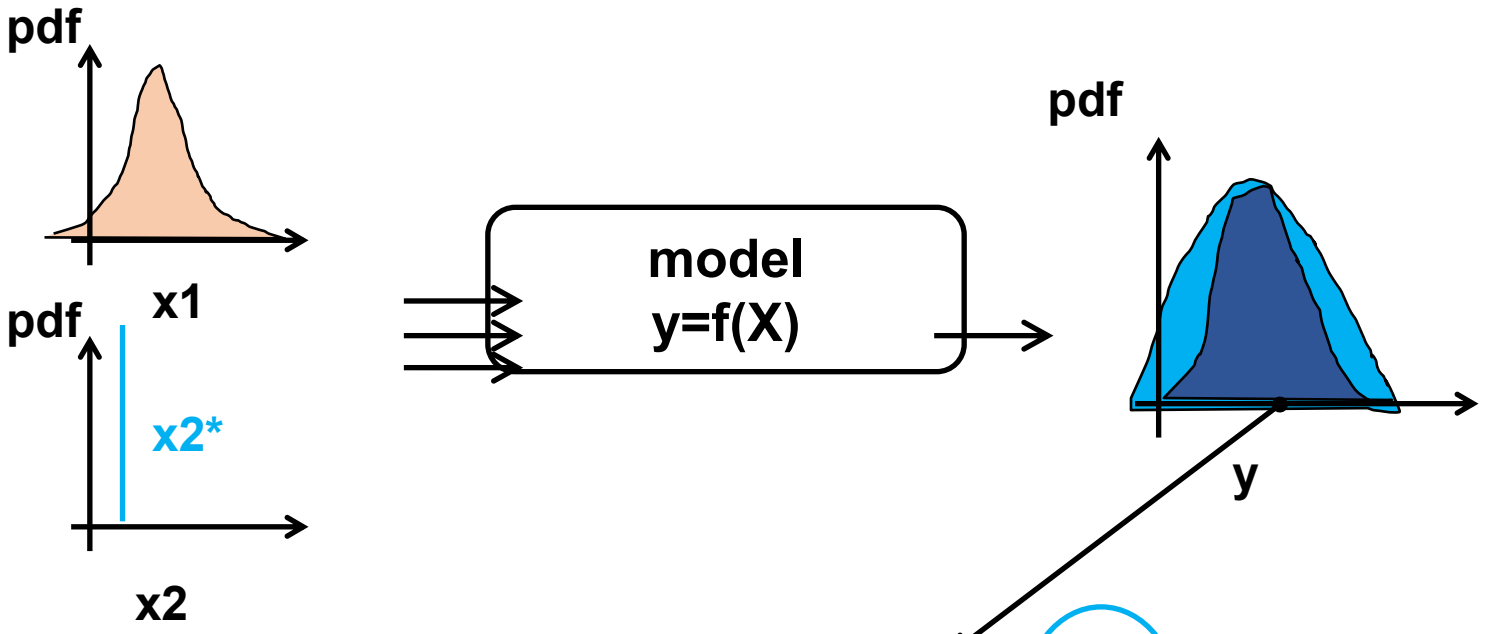
If the true value x1* was known the variance V(Y) would be reduced by V(1)



$$V(Y|X_1 = x_1^*)$$

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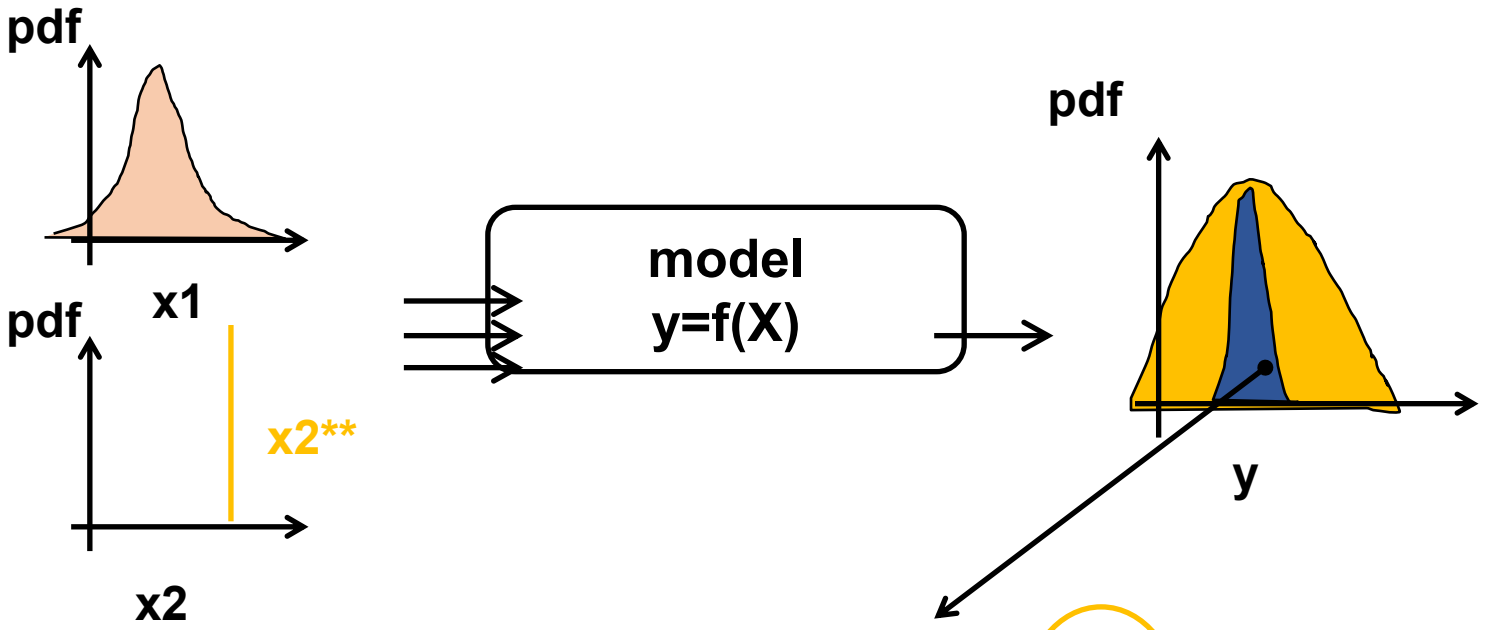
If the true value x2* was knwown the variance V(Y) would be reduced by V(2)



$$V(Y|X_2 = x_2^{**})$$

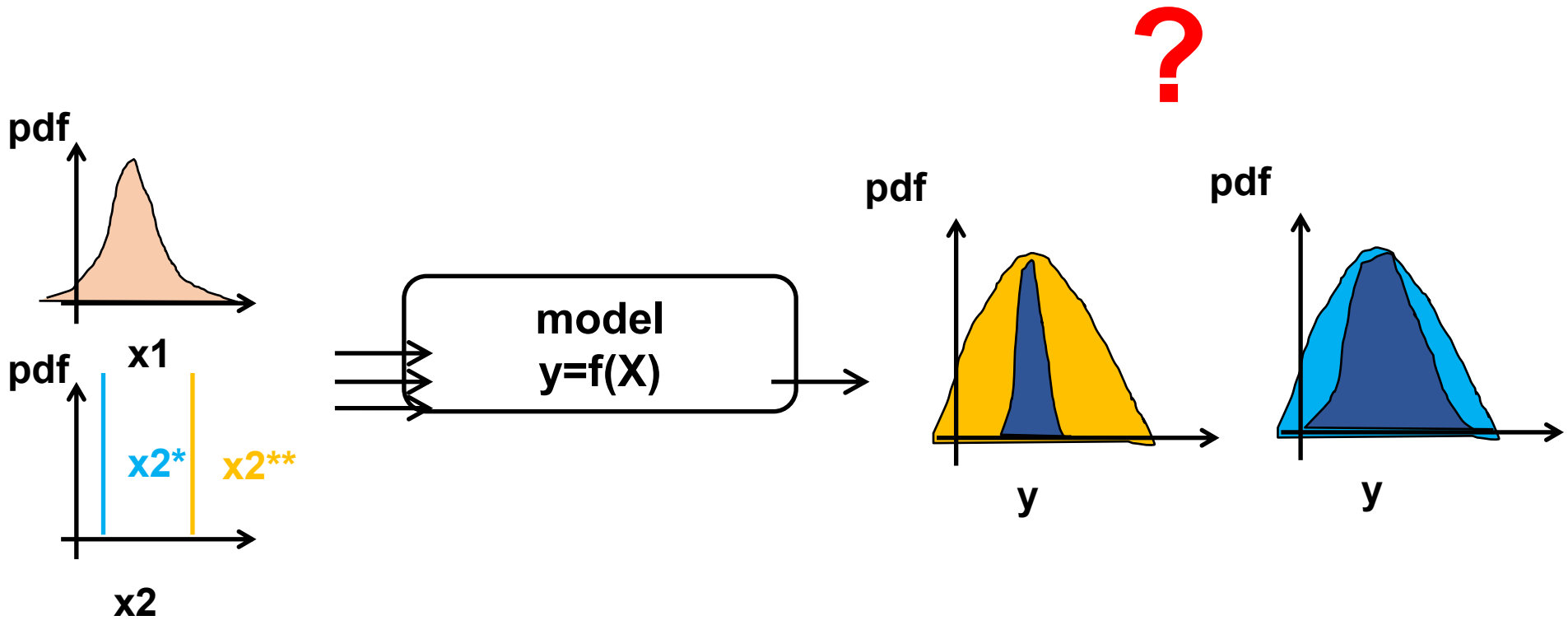
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If the true value x2** was knwown the variance V(Y) would be reduced by V(2)'



$$V(Y|X_2 = x_2^{**})$$

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Since we do not know the true value of X2

$$E(V(Y|X_2))$$

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Definition of sensitivity indices

[Saltelli et al., 2008]

Sobol' index of first order (main effect):

$$S_i = \frac{V(E(Y|X_i = x_i^*))}{V(Y)}$$

Total effect

$$S_{Ti} = 1 - \frac{V(E(Y|X_{-i}))}{V(Y)}$$

Main effects + interactions

with $X_{-i}=(X_1,\ldots, X_{i-1}, X_{i+1},\ldots X_d)$

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Use for factors' ranking and fixing

[Saltelli et al., 2008]

$$0 \le S_i \le S_{Ti} \le 1$$

- $S_{Ti} - S_i$: **interaction betw.** X_i **and all other inputs**

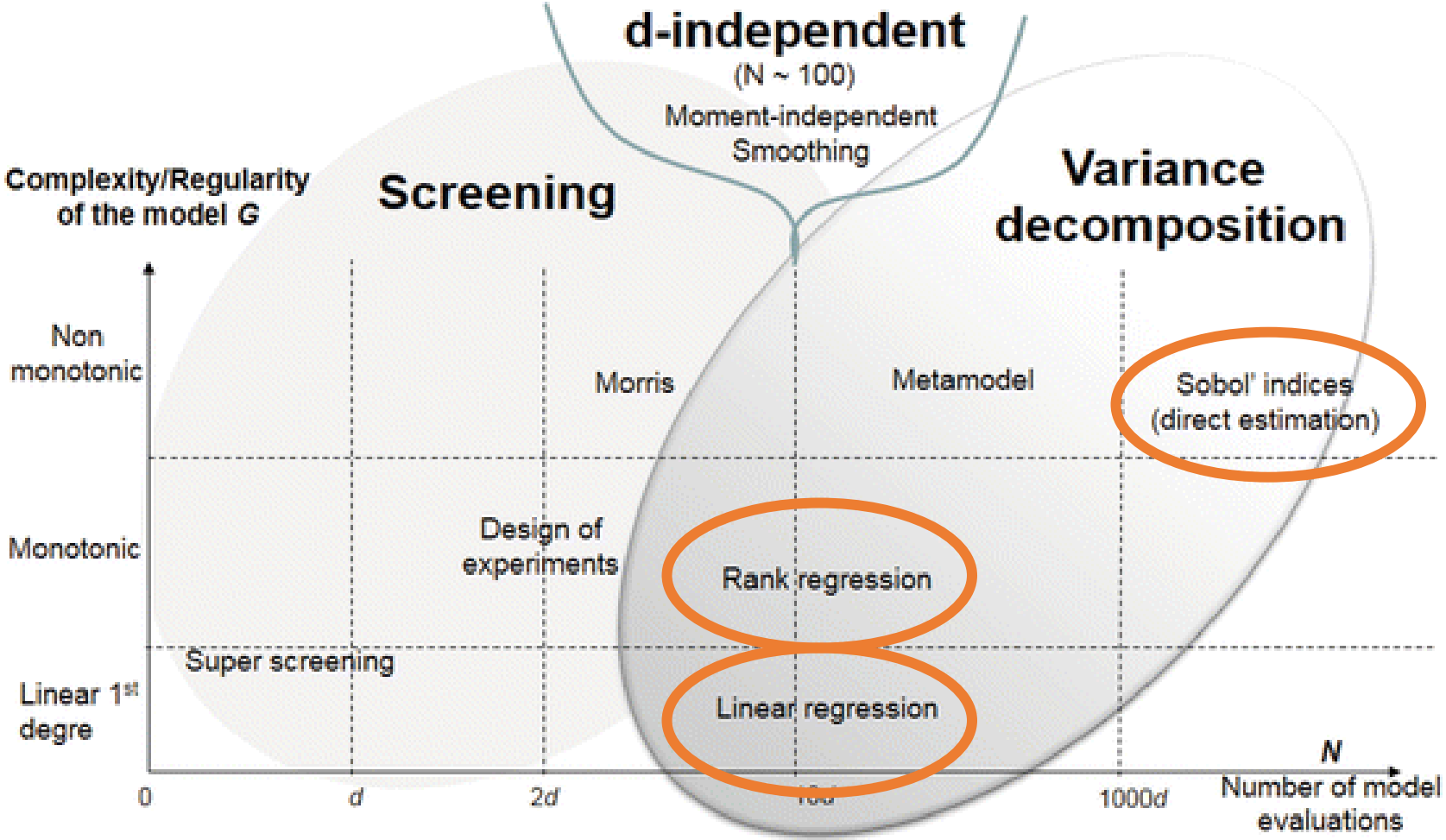
- **In practice**

  - $S_{Ti} = 0 \rightarrow f(X)$ does not depend on X_i
  - $S_i = 1 \rightarrow f(X)$ only dépends on X_i
  - $S_i = S_i^{tot}$ no interaction
  - $\sum_{i=1}^{n} S_i = 1$ additive function

- **Estimation:**

  - **Monte-Carlo 'Pick-freeze' methods** [Sobol 2001; Saltelli et al. 2010] ← requires specific experimental random design
  - **Given Samples** [rank-based: Gamboa et al. 2020; da Veiga et al. (2025)] ← only uses N random samples

[looss and Lemaitre 2015]



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A few challenges

- Correlated inputs → Shapley effects [Iooss & Prieur 2019]
- Beyond variance → HSIC [da Veiga 2015]
- Complex outputs → [Gamboa et al., 2021]
- Computational burden → use of surrogate models [see 1st session]

Merci pour votre attention!
Des questions?

Contact: j.rohmer@brgm.fr

Annexes

Definition of sensitivity indices

Law of Total Variance

[Saltelli et al., 2008]

$$V(Y) = E\left(V(Y|X_i = x_i^*)\right) + V(E(Y|X_i = x_i^*))$$

Sobol' index of first order (main effect):

$$S_i = \frac{V(E(Y|X_i = x_i^*))}{V(Y)}$$

Functional ANOVA

$$V(Y) = \sum_i V_i + \sum_{i<j} V_{ij} + \sum_{i<j<l} V_{ijl} + \ldots + V_{ij\ldots d}$$

2^d-1 terms !!

Total effect

$$S_{Ti} = 1 - \frac{V(E(Y|X_{-i}))}{V(Y)}$$

Main effects + interactions

with $X_{-i}=(X_1,\ldots, X_{i-1}, X_{i+1},\ldots X_d)$

Huge computation time cost of a single run

Versus

Large number of model runs needed

Key ingredient:
High-Performance Computing



Huge computation time cost of a single run

Versus

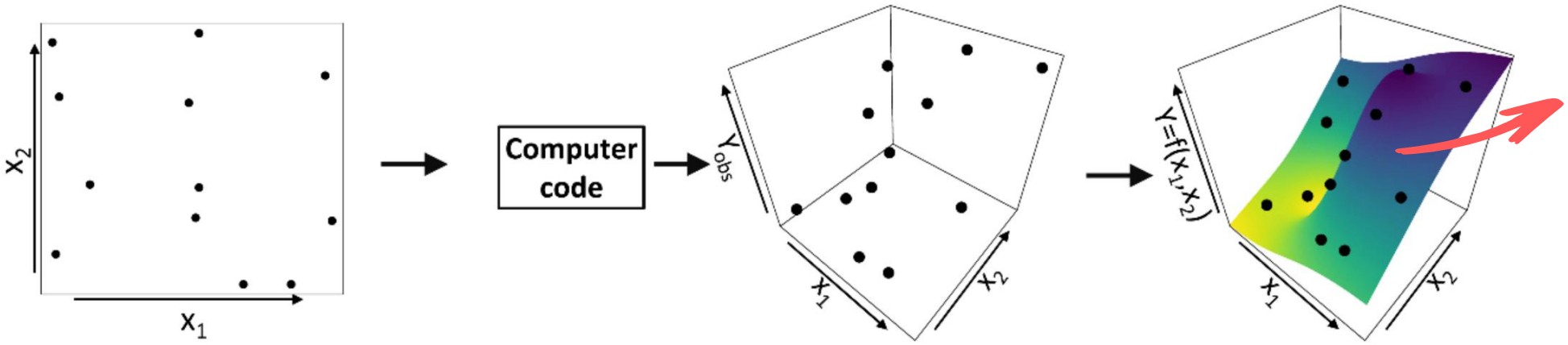
Large number of model runs needed

Complementary ingredient:
Surrogate modelling
(aka metamodelling)

Design of Experiments (DoE)
Points to perform simulations

Simulation
Performing simulations for the points of the DoE

Metamodeling
Approximation of the computer code over the Xi space



Predictive model learned using machine learning methods

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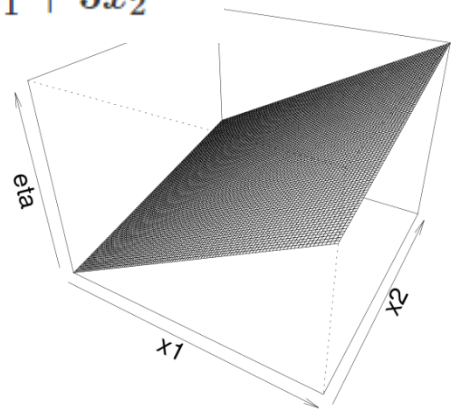
Multiple methods

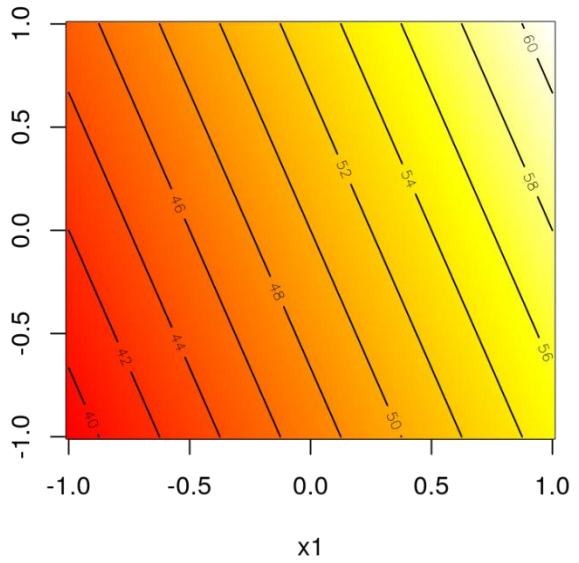
- Polynomial regression
- Smoothing techniques
- Kriging, aka Gaussian Process
- Polynomial Chaos expansion
- Support Vector Machine
- Neural network
- **All techniques of data-science / machine learning**
- Etc.

- **Validity?**
  - Comparison to the results of test set not used for training
  - In practice, cross-validation (LOO or K-fold)

Notion d'interactions

$$\eta = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$
$$= 50 + 8x_1 + 3x_2$$





$$\eta = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$$
$$= 50 + 8x_1 + 3x_2 - 4x_1 x_2$$

